



Modulation of Rhizosphere and Soil–plant–water Relations by Silicon and Zinc Improves Nutrient Acquisition in Drought-stressed Wheat

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Abstract

Drought stress reduces root hydraulic conductivity and disrupts root–soil contact, impairing rhizosphere function and plant productivity. This study evaluated the individual and combined effects of silicon (Si) and zinc (Zn) on nutrient uptake, rhizosphere chemistry, and root morphological traits in wheat (*Triticum aestivum* L.) under progressive soil drying. A factorial greenhouse experiment was conducted with three Si levels (0, 150, 300 mg kg⁻¹ as monosilicic acid), two Zn levels (0, 3 mg kg⁻¹), and two soil moisture regimes (0.4 and 0.8 field capacity, FC). Measurements included leaf and soil nutrient concentrations, rhizosphere pH, plant water status, and root traits such as rhizosheath development and root hair length and density. Drought (0.4 FC) significantly decreased leaf concentrations of phosphorus (–16.97%), potassium (–19.33%), sulfur (–7.24%), iron (–9.08%), zinc (–11.04%), copper (–25.83%), and manganese (–17.45%) compared with well-watered conditions. Application of Si or Zn mitigated these reductions in a nutrient-specific manner by enhancing rhizosphere nutrient availability, water uptake, and root functional traits. Enhanced rhizosheath formation, longer and denser root hairs, and increased carbon-rich rhizodeposition were associated with increased nutrient availability and uptake during drought. Both elements also modulated rhizosphere pH and leaf water content. Our study revealed that the application of Si and Zn in drought-stressed wheat improved nutrient acquisition. This effect was achieved through alterations in rhizosphere chemistry and root traits, particularly by enhancing rhizosheath formation and root hair development. These changes increased nutrient availability in the rhizosphere, sustained water uptake, and improved leaf water status under limited soil moisture.

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where it enhances drought tolerance through multiple mechanisms (Cheraghi et al. 2023a). Silicon deposits beneath the leaf epidermis form a silica–cuticle layer in various grass species including tall fescue (*Festuca arundinacea*) and rice (*Oryza sativa*), which diminishes cuticular transpiration and limits water loss (Vandeger et al. 2021; Yoshida et al. 1962). It facilitates the buildup of compatible solutes, including proline and sugars, which support leaf water retention and help prevent stomatal closure as demonstrated in wheat (Ashfaq et al. 2022). Silicon additionally influences stomatal function, mitigating the generation of reactive oxygen species (ROS) and alleviating oxidative stress during drought conditions (Cheraghi et al. 2024; Mostofa et al. 2021). Furthermore, Si enhances both the density and length of root hairs and stimulates rhizosheath development in wheat, strengthening root–soil interactions and promoting water and nutrient uptake under soil drying conditions (Cheraghi et al. 2024).

Zinc plays crucial roles in enzyme function, oxidative stress mitigation, and auxin-mediated root development in a wide range of crops including wheat and rice (Broadley et al. 2007; Castiglione et al. 2023). Adequate Zn improves stomatal conductance and osmotic balance, reducing drought-induced ROS accumulation and cellular damage in various crops including wheat and quinoa (*Chenopodium quinoa*) (Sattar et al. 2021; Zahmatkesh et al. 2024). Zinc influences tryptophan synthesis, a precursor to auxin, which stimulates root cell division and root hair formation, enhancing root surface area (Castillo-González et al. 2018; Roychoudhry and Kepinski 2022). These effects collectively improve plant water relations and nutrient acquisition during drought. However, while both Si and Zn improve drought tolerance via morphological and physiological mechanisms, their specific effects on rhizosphere chemistry and nutrient uptake remain underexplored.

This study aims to elucidate how Si and Zn applications modulate nutrient uptake by altering rhizosphere chemical properties, root morphological traits, and plant–soil–water interactions in wheat under varying soil moisture regimes. We hypothesize that drought-induced limitations in nutrient uptake primarily result from disrupted root–soil contact and decreased nutrient bioavailability due to fixation and precipitation in drying soil. The application of Si or Zn is expected to enhance rhizosphere nutrient availability by influencing root hair development, promoting rhizosheath formation, and regulating rhizosphere pH, while also improving plant water relations. These combined effects are anticipated to alleviate drought-induced nutrient limitations and support wheat growth under water-deficit conditions.

2 Materials and Methods

2.1 Soil Preparation and Characterization

Topsoil was collected from a 0–30 cm depth of a research farm located in Karaj, Iran (35° 45' 49.9" N, 50° 57' 0.7" E). The native soil was characterized by inherently low fertility, particularly with respect to Si and Zn, making it suitable for nutrient amendment studies. The collected soil was air-dried, lightly ground, and passed through a 4 mm sieve for subsequent pot experiments. For chemical and physical analyses, a subsample was further sieved using a 2 mm mesh. Textural analysis classified the soil as sandy loam (59% sand, 28% silt, 13% clay). The soil had a pH of 7.5, electrical conductivity (EC) of 1.12 dS m⁻¹, and contained 0.498% organic carbon (OC). The field capacity (FC) was measured at 23.5% (w/w). The concentrations of plant-available nutrients were measured as follows: Olsen phosphorus (P), 3.5 mg kg⁻¹; potassium (K), 160 mg kg⁻¹; and DTPA-extractable micronutrients—manganese (Mn), copper (Cu), zinc (Zn), and iron (Fe)—at 6.5, 0.40, 0.66, and 4.2 mg kg⁻¹, respectively (Dane and Topp 2020; Sparks et al. 2020). Plant-available silicon was measured at 29.03 mg kg⁻¹ using sequential extraction with 0.01 M CaCl₂ followed by 0.5 M acetic acid (Georgiadis et al. 2017; Korndörfer et al. 1999).

2.2 Application of Si and Zn Treatments and Soil Incubation

The experiment included three main treatment factors: (i) Si application, (ii) Zn application, and (iii) contrasting soil water content (SWC). Silicon treatments were applied at 0, 150, and 300 mg Si kg⁻¹ of soil using monosilicic acid (H₄SiO₄) as the source, while zinc was supplied at 0 and 3 mg Zn kg⁻¹ of soil in the form of zinc sulfate (ZnSO₄·7 H₂O). The treatments were applied both individually and in combination to assess their effects under contrasting moisture conditions. Soil moisture was controlled at two levels: optimal (0.7–0.8 field capacity, FC) and drought stress (0.3–0.4 FC), simulating well-watered and soil drying conditions, respectively. Prior to the application of Si and Zn, macro- and micronutrient contents were adjusted to optimal levels to eliminate confounding nutrient deficiencies. To provide essential nutrients for plant growth, the soil was enriched with K (200 mg kg⁻¹, as K₂SO₄), P (20 mg kg⁻¹, as (NH₄)₂HPO₄), and nitrogen (50 mg kg⁻¹, as (NH₄)₂CO₃). Minor elements were also added, including 0.5 mg kg⁻¹ of Fe and Cu, and 6 mg kg⁻¹ of Mn, supplied as FeSO₄·7 H₂O, CuSO₄·5 H₂O, and MnSO₄·H₂O, respectively.

Following nutrient balancing, the soil was allocated into six treatment groups according to the different combinations

of Si and Zn. Treatments were applied by spraying the designated Si and Zn solutions evenly onto the dry soil, followed by thorough homogenization through successive mixing steps to ensure uniform distribution. The amended soils were maintained at 23–27 °C for a 30-day incubation period (Wu et al. 2015). Soil moisture during incubation was maintained at field capacity to facilitate equilibrium between added nutrients and the soil matrix. Additionally, three controlled drying cycles were imposed to simulate natural wet–dry fluctuations. Every six days, soils were removed from the pots, thoroughly mixed to maintain treatment uniformity, and returned for continued incubation.

2.3 Experimental Setup: Rhizosphere–bulk Soil Separation, Plant Cultivation, and Drought Stress Application

A greenhouse study was carried out to investigate how Si and Zn interact to influence nutrient availability in the rhizosphere and their uptake by wheat under varying soil moisture conditions. The experiment was arranged as a completely randomized factorial design, with three replicates for each treatment. Wheat (*Triticum aestivum* L., cv. *Ehsan*) was grown in cylindrical plastic pots 21 cm high and 11 cm in diameter, each containing a rhizobag placed at the center measuring 10 cm in height and 7 cm in diameter. The rhizobag featured a 60 µm nylon mesh, which allowed the movement of water and nutrients but restricted root penetration, effectively enabling spatial separation between rhizosphere and bulk soil compartments.

A total of 1200 g of treated and incubated soil was added to each pot, with 200 g allocated to the rhizobag to simulate rhizosphere soil and the remaining 1000 g distributed as bulk (non–rhizosphere) soil surrounding the rhizobag. The soil surface inside the rhizobag was leveled to match that of the surrounding bulk soil to ensure uniform water distribution and root development potential. Seven pre-germinated seeds were sown within each rhizobag. During the establishment phase, soil moisture was maintained daily at 70–80% of FC by gravimetric adjustment. Drought treatments were applied 20 days after planting, once the wheat plants had developed adequate root and shoot biomass. Two contrasting irrigation regimes were imposed: (i) well-watered conditions maintained at 70–80% of FC, and (ii) drought stress maintained at 30–40% of FC. Soil moisture levels were controlled by daily weighing and water replenishment to the desired FC. The wheat plants were allowed to grow, and the drought cycles were maintained, up to 45 days after sowing. This design enabled precise assessment of rhizosphere responses and plant nutrient uptake under combined Si, Zn, and SWC treatments.

2.4 Measurements

2.4.1 Plant Transpiration and Relative Leaf Water Content (RWC)

To monitor the physiological response of wheat plants to drought, a soil drying cycle was initiated when the plants reached 30 days of age and attained a height of 25–30 cm, corresponding to 10 days after the onset of drought stress. Prior to this, SWC in all pots was standardized to 80% of field capacity (0.8 FC) to allow natural depletion of water through transpiration. Transpiration was measured gravimetrically by weighing each pot twice daily, at 9:30 in the morning and 4:30 in the afternoon. Daytime transpiration was calculated by dividing the weight difference between the two daily measurements by the time interval (Cheraghi et al. 2024). Water loss from the soil surface was considered negligible because a 3 cm layer of perlite covered the soil.

Relative leaf water content was measured to assess the water status of plants under drought conditions. Fully expanded upper leaves (second or third from the apex) were sampled from each treatment at 43–45 days after sowing. Fresh weight (FW) was measured immediately, after which they were submerged in distilled water for 2 h to achieve full turgidity. After carefully blotting off surface water, the turgid weight (TW) of the samples was measured. The samples were subsequently oven-dried at 65 °C until their dry weight (DW) stabilized. RWC was determined using the procedure outlined by Turner (1981). Data were collected for three successive days during the drought cycle, and average values were employed in the statistical (Eq. 1).

$$\text{RWC (\%)} = \frac{\text{FW} - \text{DW}}{\text{TW} - \text{DW}} \times 100 \quad (1)$$

2.4.2 Determination of Nutrient Content in Wheat Leaves

Leaf tissue nutrient concentrations were analyzed to evaluate the impact of Si and Zn application under drought conditions. The dried leaf samples were processed into a fine powder with a stainless-steel grinder. A 0.5 g sample of the ground leaf tissue was subjected to digestion in a closed-vessel microwave system for multielement analysis (Anton Paar, Austria). Samples were treated with 5 mL of 65% nitric acid and 2 mL of 30% hydrogen peroxide, allowing a 30-minute pre-reaction at room temperature before proceeding with microwave digestion. The concentrations of K, P, calcium (Ca), magnesium (Mg), sulfur (S), Fe, Mn, Zn, Cu, and boron (B) in the final solution were measured

using Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES; Optima 2100 DV, PerkinElmer, USA).

Nitrogen (N) content was determined using the Kjeldahl method. In brief, the plant material was digested with concentrated sulfuric acid and a catalyst mixture to convert nitrogen into ammonium sulfate. Following alkalization with sodium hydroxide, ammonia evolved from the sample was absorbed into a boric acid solution. The ammonia absorbed in boric acid was titrated with 0.01 N sulfuric acid, and total nitrogen content was determined from the amount of acid used (Horneck and Miller 2019). To measure silicon concentration, 25 mg of powdered leaf material was digested in specialized TFM microwave vessels using 5 mL of 40% hydrofluoric acid (HF) and 3 mL of aqua regia. The digestion procedure was performed at temperatures reaching 250 °C and pressures up to 8 bar. After cooling, the digest was diluted to 100 mL with deionized water, and the Si concentration was measured by ICP-OES at the characteristic wavelength of 251.611 nm, as described by Neumann et al. (2011).

2.4.3 Quantification of Rhizosheath Formation

Rhizosheath formation was quantified at the end of the experimental period, when SWC had declined to approximately 0.3 of FC. To preserve the rhizosheath structure, each pot was carefully cut open along the sides, and the root system was gently removed with minimal disturbance. Loose soil particles were carefully shaken off, and the soil that remained firmly attached to the roots was defined as the rhizosheath. Soil tightly adhering to the roots was manually removed and placed into containers that had been weighed beforehand. To determine rhizosheath dry weight, the collected soil samples were dried in an oven at 105 °C for 24 h (Delhaize et al. 2012). To ensure accuracy, only root segments with visibly preserved rhizosheath were used in the measurement. After removing the rhizosheath, the root segments were carefully washed with distilled water, and both fresh and dry weights were measured. The root samples were oven-dried under the same conditions (105 °C for 24 h) to obtain their dry weight. The amount of rhizosheath was expressed as grams of soil dry weight attached per gram of root dry weight.

2.4.4 Measurement of PH and OC in Rhizospheric and Bulk Soil

Soil pH in both bulk (non-rhizospheric) and rhizospheric soils was determined through the saturated paste method following the procedure of Rhoades (1996). For this purpose, a paste was prepared by adding distilled water to air-dried, 2 mm-sieved soil until saturation was achieved.

After the paste reached equilibrium, pH was recorded using a properly calibrated digital pH meter. Soil organic carbon was determined by applying the dichromate oxidation procedure developed by (Walkley and Black 1934). One gram of oven-dried soil was treated with 10 mL of 1 N $K_2Cr_2O_7$ solution and 20 mL of concentrated H_2SO_4 for digestion. Following a 30-minute reaction period at ambient temperature, the excess dichromate was titrated against 0.5 N ferrous sulfate ($FeSO_4$) using ferroin as an indicator. Each analysis was carried out in triplicate, and the outcomes were reported as percentage values of organic carbon.

2.4.5 Determination of Nutrient Element Concentrations in Rhizospheric and Bulk Soil

Available P and K in rhizospheric and bulk soil samples were determined using the Olsen (1954) and ammonium acetate extraction methods, respectively. To determine P, soil samples were treated with 0.5 M $NaHCO_3$ (pH 8.5) as the extracting agent, and the absorbance was recorded at 882 nm with a spectrophotometer. Soil K was obtained by extraction with 1 N ammonium acetate and subsequently quantified by means of a Corning 410 flame photometer, as described by (Shahbazi et al. 2022). Micronutrients (Fe, Zn, Cu, and Mn) extractable with Diethylene Triamine Pentaacetic Acid (DTPA) were obtained following the procedure outlined by Sparks et al. (2020) and their concentrations were determined using atomic absorption spectroscopy (AAS, PinAAcle 900 F). To evaluate plant-available Si, a modified sequential extraction protocol for alkaline soils was applied (Georgiadis et al. 2017). The method consisted of an initial extraction with 0.01 M $CaCl_2$ to release soluble Si, followed by 0.5 M acetic acid for the adsorbed fraction. Both extracts were quantified using ICP-OES, and the sum of these values was reported as the plant-available Si.

2.4.6 Investigation of Si and Zn Effects On Root Hair Growth in a Hydroponic System

The effects of Si and Zn on wheat root hair development were evaluated using a controlled hydroponic system under greenhouse conditions. Wheat seeds were initially germinated in Petri dishes on moistened filter paper with either distilled water (control), 2 mM monosilicic acid (Si treatment), or 2 mM zinc sulfate (Zn treatment). About 3 days after the appearance of the radicle, uniform seedlings were carefully transferred into cylindrical glass containers with a height of 9.5 cm and a diameter of 2.5 cm, each filled in advance with 50 mL of the respective treatment solution. To establish a mild drought condition, polyethylene glycol (PEG) 6000 was added at a concentration of 20% (W/V) to the nutrient solution, lowering its osmotic potential to

−0.7 MPa. Control plants, in contrast, were supplied with distilled water to maintain favorable water availability. The experimental treatments were organized according to a completely randomized design, including three replications for each treatment under both stress and non-stress conditions. Over a period of 10 days, the plants were maintained under a photoperiod consisting of 12 h of light and 12 h of darkness. The temperature was regulated at 18 °C during the day and 22 °C at night, while relative humidity was kept at 65% according to (Cheraghi et al. 2024).

When the growth period ended, seedlings were carefully extracted from the hydroponic medium to prevent any harm to the roots. Root hair measurements were performed on a 2–3 cm segment from the root apex. Immediately after extraction, the roots were examined using a stereomicroscope fitted with a digital camera. The length of root hairs was determined by manually tracing each hair on high-resolution images using ImageJ software at a standardized magnification. The density of root hairs was assessed by enumerating the hairs per millimeter of root length within identical segments, based on digital images analyzed with ImageJ software.

2.5 Data Analysis

A three-factor analysis of variance was performed to evaluate both the individual and interactive impacts of Si, Zn, and soil moisture on all recorded parameters. To assess the effect of the rhizosphere independently of treatment combinations, a one-way ANOVA was conducted using a completely randomized design. In the hydroponic experiment, a completely randomized design was applied to analyze the effects of Si and Zn on root hair length and density under both optimal and drought stress conditions. Duncan's multiple range test was employed to compare means, with statistical significance set at $p \leq 0.05$. All statistical analyses were performed with SAS software version 9.4, and graphs were generated using Microsoft Excel 2016. Data are expressed as mean values with their corresponding standard deviations, based on three biological replicates per treatment.

3 Results

3.1 Effects of Si, Zn, and SWC On Leaf Macronutrient Concentrations

Leaf N concentration increased by 18.1% with the addition of 150 mg Si per kg of soil and by 19.4% when 300 mg Si per kg was applied (Fig. 1a). Zinc application at 3 mg kg^{−1} enhanced leaf N by 10.8% (Fig. 1b), while soil moisture levels had no significant effect on N concentration. In

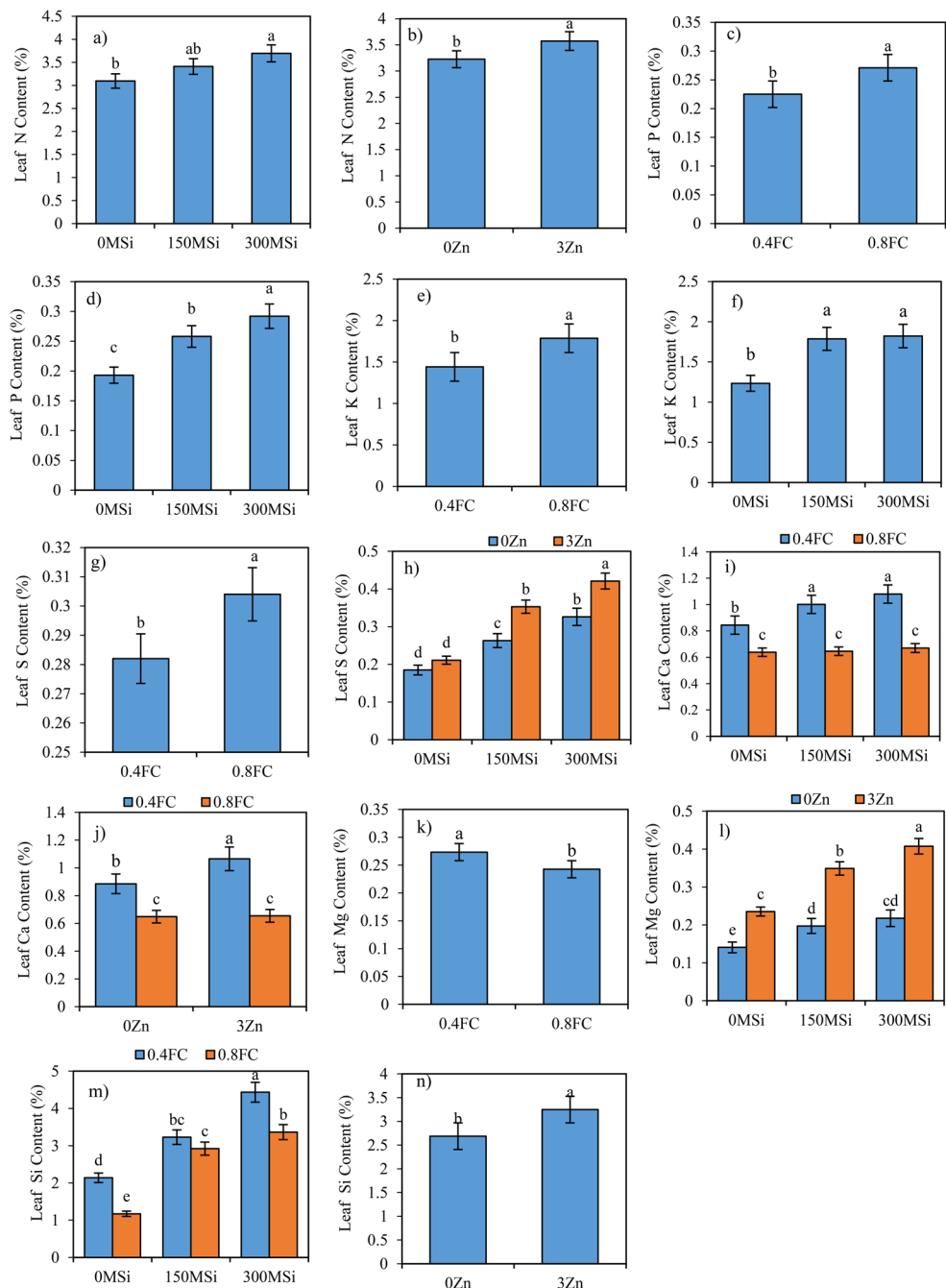
contrast, decreasing soil moisture from 0.8 to 0.4 FC caused a significant 17.0% reduction in leaf P levels (Fig. 1c). Silicon application significantly improved P concentration, increasing it by 33.7% and 51.3% at 150 and 300 mg Si kg^{−1}, respectively (Fig. 1d). Drought stress also led to a 19.3% decrease in leaf K concentration (Fig. 1e), which was substantially mitigated by Si, raising K levels by approximately 45–48% at both application rates (Fig. 1f). Leaf S concentration decreased by 7.2% under drought (Fig. 1g). However, combined application of Si and Zn significantly elevated S levels; co-application of Zn (3 mg kg^{−1}) with 150 and 300 mg Si kg^{−1} increased S concentrations by 34.2% and 29.1%, respectively, relative to Si alone, and by 67.3% and 99.5% relative to Zn alone (Fig. 1h).

Under drought conditions, Si increased leaf Ca concentrations by 18.6% and 27.9% at 150 and 300 mg Si kg^{−1}, respectively, whereas no significant effects were observed under optimal moisture (Fig. 1i). Zinc improved Ca content by 20.3% under drought but had no effect under well-watered conditions (Fig. 1j). Interestingly, drought stress alone increased Mg concentration by 12.7% (Fig. 1k). Co-application of Si and Zn further augmented Mg accumulation, enhancing Mg levels by 48.3% and 73.2% at 150 and 300 mg Si kg^{−1}, respectively, compared to Zn treatment alone (Fig. 1l). Leaf Si concentration was significantly influenced by both Si supplementation and moisture regime (Table 1). Compared with plants under well-watered conditions, drought stress increased Si content by 82.6%, 10.5%, and 31.8% in the control, 150 mg Si kg^{−1}, and 300 mg Si kg^{−1} treatments, respectively (Fig. 1m). Additionally, Zn application elevated leaf Si concentration by 20.9% compared to untreated controls (Fig. 1n).

3.2 Effects of Si, Zn, and SWC on Leaf Micronutrient Concentrations

Exposure to drought stress reduced leaf Fe concentration by 9.08%. Leaf Fe concentration was significantly affected by the combined effects of Si application and SWC. Application of 150 and 300 mg Si per kg enhanced Fe levels by 26.71 and 41.22% under drought stress, and by 43.12 and 12.40% under well-watered conditions, respectively (Fig. 2a). Zinc application alone also enhanced leaf Fe concentration by 29.37% (Fig. 2b). Decreasing soil moisture resulted in an 11.04% reduction in leaf Zn concentration (Fig. 2c). However, combined Si and Zn treatments markedly elevated Zn content (Table 1; Fig. 2d). In the absence of Zn, Si at 150 and 300 mg kg^{−1} increased leaf Zn concentration by 52.87% and 73.53%, respectively. When Zn (3 mg kg^{−1}) was co-applied, Zn concentration increased by 155.83%, 64.00%, and 59.77% for 0, 150, and 300 mg Si kg^{−1}, respectively, compared to the corresponding Si-only treatments.

Fig. 1 Effects of soil water content (SWC), silicon (Si), and zinc (Zn) treatments on macronutrient concentrations in wheat leaves. Subpanels display concentrations of nitrogen (**a**, **b**), phosphorus (**c**, **d**), potassium (**e**, **f**), sulfur (**g**, **h**), calcium (**i**, **j**), magnesium (**k**, **l**), and silicon (**m**, **n**) across treatments. The treatments consisted of 0Si with no Si addition, 150MSi and 300MSi corresponding to 150 and 300 mg of Si per kg of soil, 0Zn with no Zn addition, and 3Zn equivalent to 3 mg of Zn per kg of soil, under two SWC: 0.4 field capacity (FC) representing drought stress, and 0.8 FC representing optimal moisture. Treatments marked with different letters differ significantly according to Duncan's multiple range test, with significance set at $p \leq 0.05$



Leaf Cu concentration declined by 25.83% under drought (Fig. 2e). Application of Si at 150 and 300 mg kg⁻¹ elevated Cu levels by 8.94% and 17.51%, respectively (Fig. 2f), while Zn supplementation increased Cu by 11.33% relative to the control (Fig. 2g). The concentrations of Mn and B were significantly influenced by the combined effects of Zn application and soil moisture levels. Mn concentration was consistently higher under optimal conditions compared to drought. Zn application had no significant effect on Mn under drought but reduced Mn concentration by 9.86% under optimal moisture (Fig. 2h). In contrast, Si

significantly increased Mn concentration in a dose-dependent manner (Fig. 2i). Interestingly, boron concentration in the leaves was elevated under drought stress compared to well-watered conditions. While Zn had no effect on B under optimal moisture, it increased B concentration by 28.77% under drought stress (Fig. 2j). Similarly, Si application at 150 and 300 mg kg⁻¹ enhanced leaf B by 10.23% and 19.01%, respectively (Fig. 2k).

Table 1 NOVA results for the effects of soil water content (SWC), silicon (Si), and zinc (Zn) treatments on macro- and micronutrient concentrations in wheat leaves

S.OV	df	Mean square						
		Leaf N Content (%)	Leaf P Content (%)	Leaf K Content (%)	Leaf S Content (%)	Leaf Ca Content (%)	Leaf Mg Content (%)	Leaf Si Content (%)
Si	2	1.081**	0.031**	1.306**	0.0941**	0.0508*	0.0485**	11.79**
Zn	1	1.082*	0.0019ns	0.0046ns	0.0442**	0.0783*	0.1901**	2.83**
SWC	1	0.016ns	0.0187**	1.074**	0.0043*	0.9444**	0.0084**	3.25**
Si × Zn	2	0.130ns	0.0001ns	0.0006ns	0.0042*	0.0578ns	0.0069**	0.0072ns
Si × SWC	2	0.079ns	0.0003ns	0.0169ns	0.0015ns	0.0374*	0.00016ns	0.504*
Zn × SWC	1	0.139ns	0.00032ns	0.000031ns	0.0028ns	0.0682*	0.00034ns	0.0128ns
Si × Zn × SWC	2	0.008ns	0.00007ns	0.0027ns	0.00071ns	0.0130ns	0.00009ns	0.0432ns
Error	24	0.158	0.00055	0.023	0.00088	0.0136	0.00069	0.098
CV%	-	11.71	9.49	9.35	10.14	14.34	10.22	10.55
S.OV	df	Leaf Fe Content (mg kg ⁻¹)	Leaf Zn Content (mg kg ⁻¹)	Leaf Cu Content (mg kg ⁻¹)	Leaf Mn Content (mg kg ⁻¹)	Leaf B Content (mg kg ⁻¹)		
Si	2	605.30**	121.54**	3.45*	711.75**	20.28**		
Zn	1	2278.63**	2297.12**	4.60*	2.15ns	70.89**		
SWC	1	315.35**	90.09*	35.22**	481.07**	454.12**		
Si × Zn	2	80.83ns	54.94*	0.963ns	6.29ns	1.27ns		
Si × SWC	2	116.50*	1.88ns	0.404ns	0.652ns	0.614ns		
Zn × SWC	1	61.91ns	0.906ns	0.022ns	133.79**	31.39**		
Si × Zn × SWC	2	20.00ns	2.29ns	0.174ns	14.65ns	1.29ns		
Error	24	24.93	14.07	0.812	12.12	2.95		
CV%	-	8.03	13.86	13.51	9.11	11.46		

Si, Zn, and SWC indicate main effects of silicon, zinc, and soil water content, respectively. Si × Zn, Si × SWC, Zn × SWC, and Si × Zn × SWC indicate their interaction effects. df: degrees of freedom; CV: coefficient of variation. *, **, and ns: significant at $P \leq 0.05$, $P \leq 0.01$, and non-significant, respectively

3.3 Rhizosphere Modulation of K, P, and Si Under Experimental Treatments

In all treatments, including the control, concentrations of K, P, and Si were consistently greater in the rhizosphere compared to the bulk soil (Fig. 3). Both rhizosphere and non-rhizosphere soils showed significant declines in K concentration as soil moisture decreased (Fig. 3a). Within the rhizospheric soil, K concentration increased by 9.99% and 13.09% following the addition of 150 and 300 mg Si per kg, respectively, whereas no significant changes were observed in bulk soil K content (Fig. 3b). Drought stress caused a reduction in P concentration by 8.75% in the rhizosphere and 10.15% in the bulk soil (Fig. 3c). In contrast, P concentrations in the rhizosphere were markedly increased by Si application, showing rises of 45.50% and 66.48% at 150 and 300 mg Si per kg of soil, respectively, relative to the untreated control. Notably, rhizosphere P concentrations were 123.99%, 145.48%, and 132.51% higher than those in the corresponding bulk soils under 0, 150, and 300 mg kg⁻¹ Si treatments, respectively (Fig. 3d). Zinc application further increased P concentration in the rhizosphere, while having no significant effect on bulk soil P (Fig. 3e).

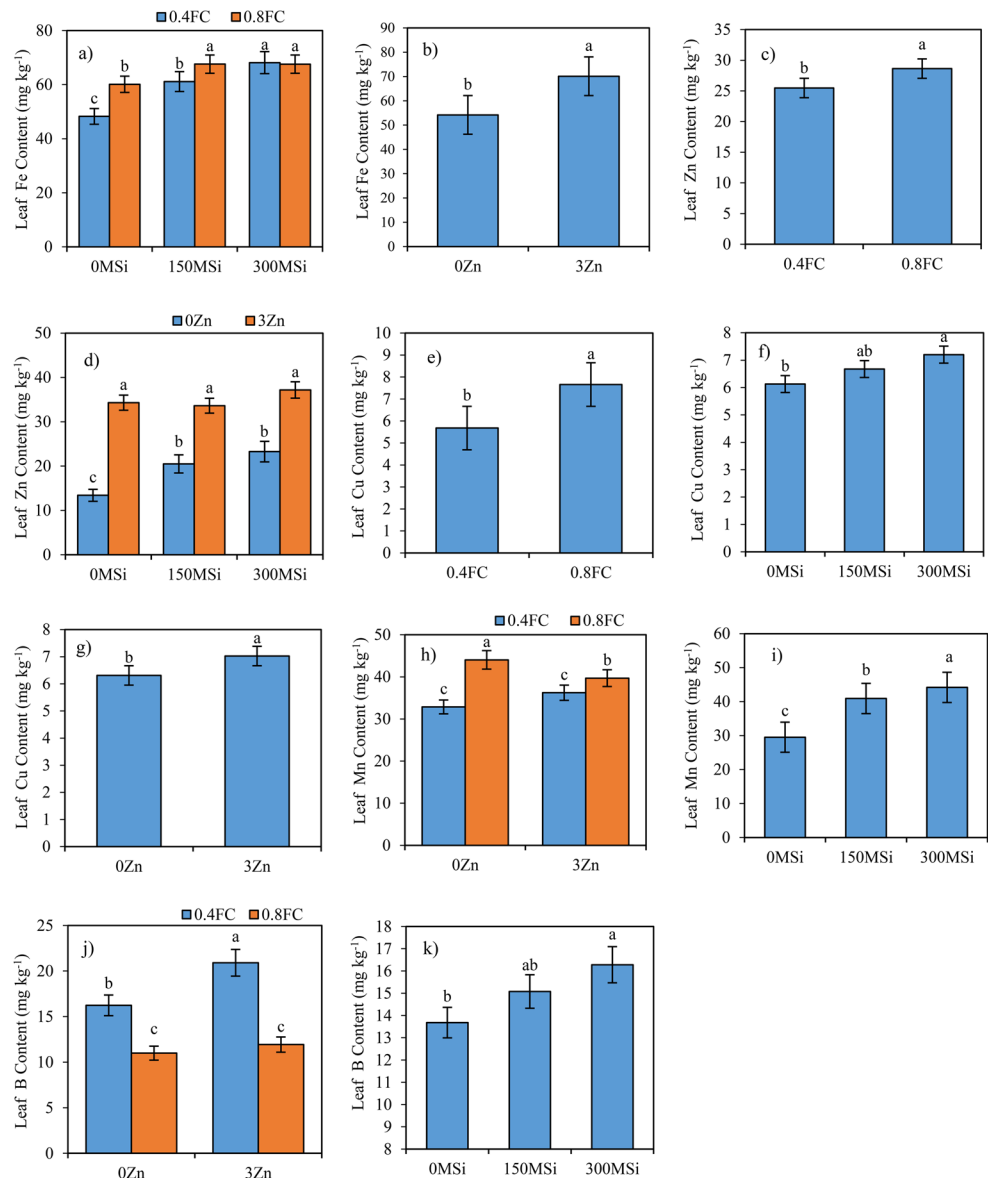
Drought stress caused a 5.17% decline in Si concentration in the bulk soil but had no significant impact on Si levels

in the rhizosphere (Fig. 3f). Zinc application increased Si concentration in the rhizosphere by 18.07%, without affecting bulk soil Si (Fig. 3g). In the absence of Si input, Si concentrations did not differ between rhizosphere and bulk soils. However, Si supplementation at 150 and 300 mg kg⁻¹ increased Si content in the rhizosphere by 24.74% and 38.24%, respectively, and in the bulk soil by 11.07% and 19.32%, respectively (Fig. 3h).

3.4 Rhizosphere Modulation of Micronutrient Availability Under Experimental Treatments

Drought stress reduced rhizospheric Fe concentration by 10.34%, while Fe levels in bulk soil remained unaffected (Fig. 4a). Applying 150 and 300 mg Si per kg of soil significantly enhanced bioavailable iron in both rhizospheric and bulk soils, while the two application levels did not differ significantly from each other (Fig. 4b). Regardless of treatment, Fe concentrations were consistently higher in rhizospheric than in bulk soil under all moisture regimes. The availability of Zn in soil was significantly influenced by soil moisture, the application of Si and Zn, as well as their interactive effects. Drought reduced bioavailable Zn by 21.25% and 7.43% in rhizospheric and bulk soils, respectively (Fig. 4c). Silicon application increased Zn concentrations in

Fig. 2 Effects of soil water content (SWC), silicon (Si), and zinc (Zn) treatments on micronutrient concentrations in wheat leaves. Subpanels display concentrations of iron (a, b), zinc (c, d), copper (e–g), manganese (h, i), and boron (j, k) across treatments. The treatments consisted of 0Si with no Si addition, 150MSi and 300MSi corresponding to 150 and 300 mg of Si per kg of soil, 0Zn with no Zn addition, and 3Zn equivalent to 3 mg of Zn per kg of soil, under two SWC: 0.4 field capacity (FC) representing drought stress, and 0.8 FC representing optimal moisture. Treatments marked with different letters differ significantly according to Duncan's multiple range test, with significance set at $p \leq 0.05$



the rhizosphere by 20.32% and 32.57% at 150 and 300 mg Si kg⁻¹, respectively, but had no significant effect on Zn levels in bulk soil (Fig. 4d). Exogenous Zn (3 mg kg⁻¹) significantly increased Zn availability in both compartments, although levels remained higher in the rhizosphere across all treatments (Fig. 4e). However, under Zn application, drought resulted in a 20.38% reduction in rhizospheric Zn content. In the absence of Zn application, Si did not alter Zn availability; in contrast, combined Si–Zn treatments markedly enhanced Zn levels in the rhizosphere but had no significant effect on bulk soil (Fig. 4f). Rhizospheric Zn concentrations increased by 37.96%, 68.67%, and 82.86% relative to bulk soil under combined 0, 150, and 300 mg Si kg⁻¹ with Zn application, respectively.

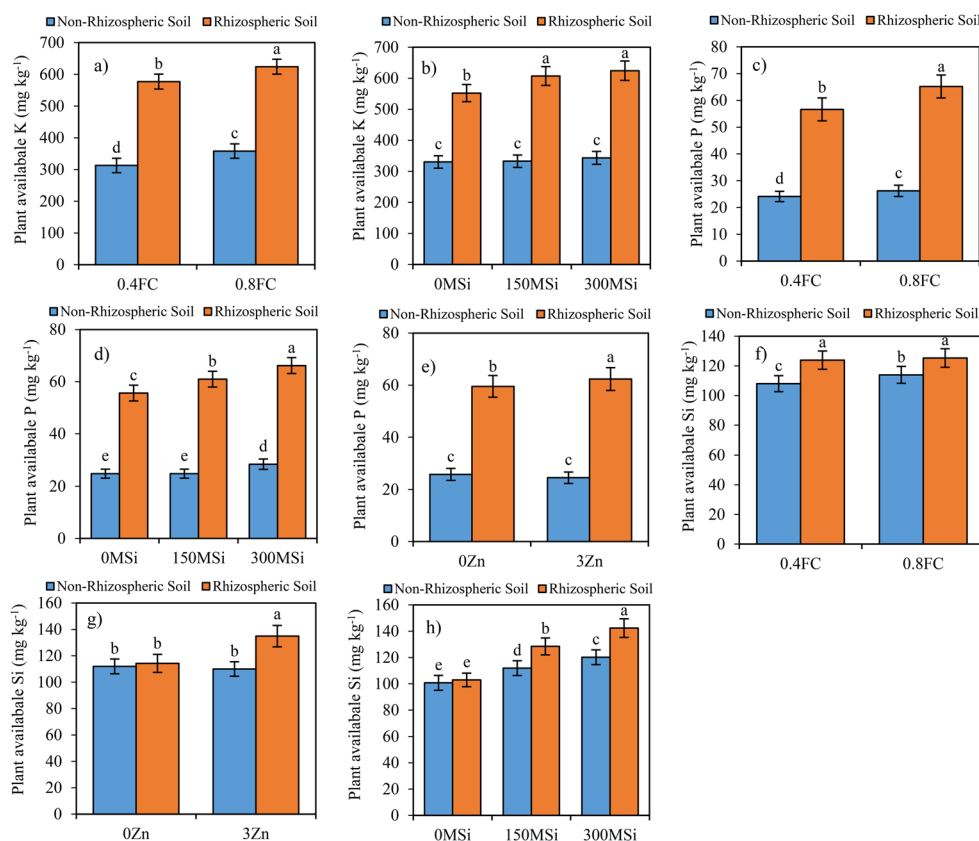
Soil Cu availability was unaffected by moisture content. However, Zn application decreased rhizospheric Cu

by 9.82% without influencing bulk soil Cu levels (Fig. 4g). Among Si treatments, only 300 mg Si kg⁻¹ significantly enhanced Cu in the rhizosphere by 15.10%, with no effect observed in the bulk soil (Fig. 4h). Zinc supplementation also reduced rhizospheric Mn by 6.95%, while Si application increased Mn levels in rhizospheric soil by 7.83% and 22.9% at 150 and 300 mg Si kg⁻¹, respectively (Fig. 4i). Across all treatments, Mn concentrations were consistently higher in the rhizosphere than in the bulk soil (Fig. 4j).

3.5 Influence of Experimental Treatments On Soil pH and OC in Rhizosphere and Bulk Compartments

Variations in soil pH and OC content under different Si, Zn, and SWC treatments are shown in Fig. 5 and Table S2. Changes in soil pH and OC content in response to Si, Zn,

Fig. 3 Effects of soil water content (SWC), silicon (Si), and zinc (Zn) treatments on potassium (a, b), phosphorus (c–e), and silicon (f–h) concentrations in rhizosphere and bulk (non-rhizosphere) soils. The treatments consisted of 0Si with no Si addition, 150MSi and 300MSi corresponding to 150 and 300 mg of Si per kg of soil, 0Zn with no Zn addition, and 3Zn equivalent to 3 mg of Zn per kg of soil, under two SWC: 0.4 field capacity (FC) representing drought stress, and 0.8 FC representing optimal moisture. Treatments marked with different letters differ significantly according to Duncan's multiple range test, with significance set at $p \leq 0.05$



and SWC treatments are presented in Fig. 5 and Table S2. Overall, rhizospheric soil exhibited significantly lower pH and higher OC compared to bulk soil. On average, the pH of rhizospheric soil was lower by 0.999 units compared to that of bulk soil (Fig. 5a). Soil water content did not significantly influence pH in either rhizospheric or bulk compartments; nonetheless, rhizosphere pH was consistently 0.083 to 0.915 units lower than that of bulk soil across both moisture conditions (Fig. 5b). Application of Si decreased rhizosphere pH by 0.357 to 0.466 units compared to the control. The pH differential between rhizosphere and bulk soil widened with increasing Si dose, reaching 1.082 and 1.167 units under 150 and 300 mg Si kg⁻¹, respectively (Fig. 5c). Zn application further reduced rhizospheric pH by 0.2899 units, with no significant change observed in the bulk soil (Fig. 5d).

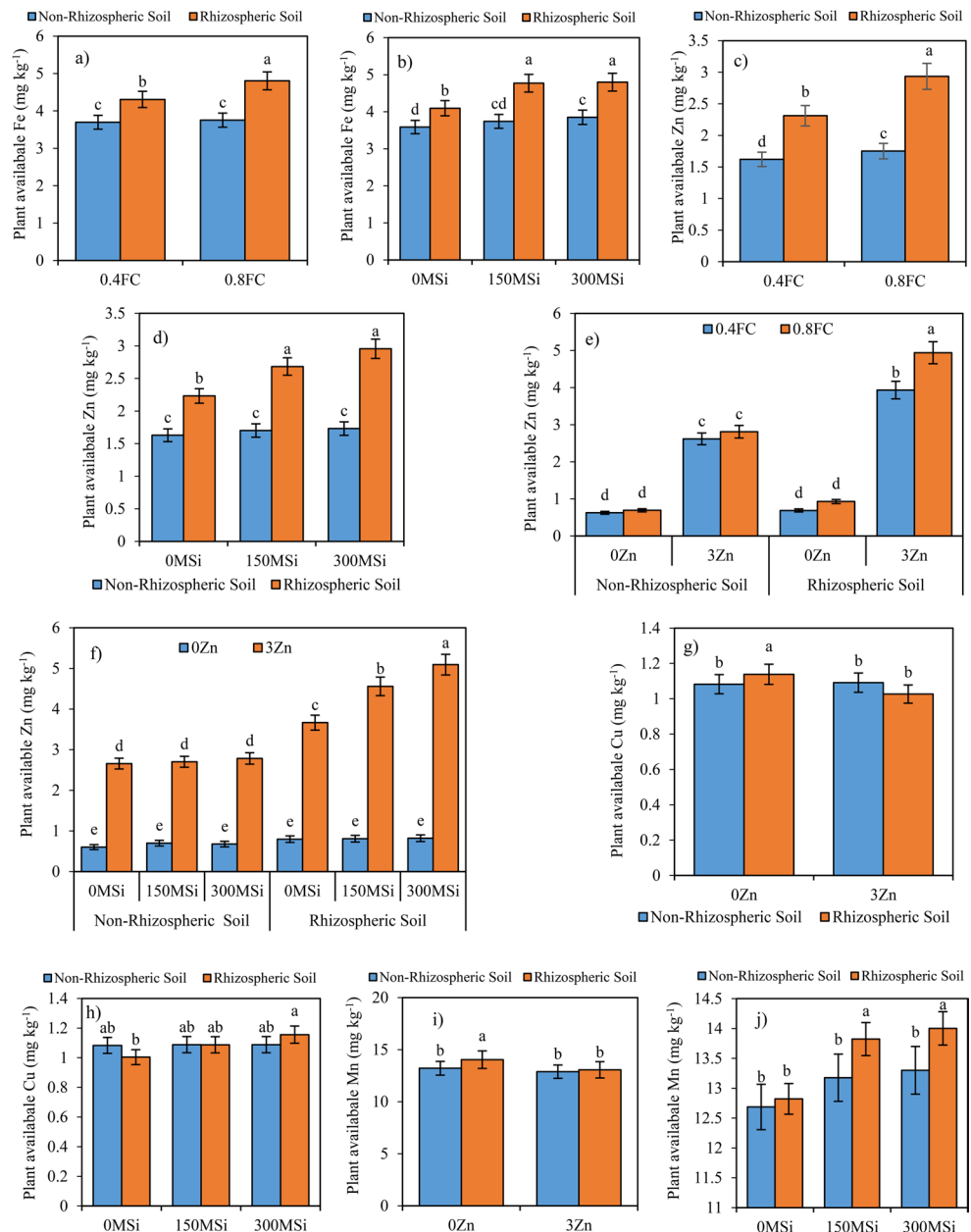
Rhizospheric OC content was markedly higher than in bulk soil, with an overall increase of 169.07% (Fig. 5e). Reducing soil moisture increased OC content by 7.58% and 17.53% in rhizospheric and bulk soil, respectively (Fig. 5f). Silicon application enhanced OC content in both compartments, with increases of 16.65% and 28.85% in the rhizosphere and 0.75% and 13.6% in bulk soil under 150 and 300 mg Si per kg of soil, respectively (Fig. 5g). Consequently, OC in the rhizosphere was 1.39, 1.77, and 1.90 times greater than that of bulk soil under control, 150, and 300 mg Si per kg, respectively. Independent Zn application

increased rhizospheric OC by 22.84% but had no significant effect in bulk soil (Fig. 5h).

3.6 Influence of Experimental Treatments On Transpiration Rate and RWC

Under drought stress and without Si application, transpiration rate (E_{norm}) was reduced by 31.07% compared to well-watered conditions. However, Si application at 150 and 300 mg kg⁻¹ increased E_{norm} by 14.12% and 32.39%, respectively, under drought, and by 1.48% and 5.88% under optimal moisture (Fig. 6a). Independent Zn application enhanced E_{norm} by 4.99% (Fig. 6b). A similar pattern was recorded for RWC. Under drought conditions, RWC declined by 22.01% without Si addition, while supplementation with 150 and 300 mg Si per kg improved RWC by 19.80% and 22.44%, respectively. Under optimal moisture, the same Si doses increased RWC by 1.33% and 3.25%, respectively (Fig. 6c). Similarly, Zn application increased RWC by 2.80% irrespective of soil moisture level (Fig. 6d). These findings suggest that both Si and Zn mitigate drought-induced declines in transpiration and leaf water status, with Si exerting a stronger dose-dependent effect under stress conditions.

Fig. 4 Effects of soil water content (SWC), silicon (Si), and zinc (Zn) treatments on iron (a, b), zinc (c–f), copper (g, h), and manganese (i, j) concentrations in rhizosphere and bulk (non-rhizosphere) soils. The treatments consisted of 0Si with no Si addition, 150MSi and 300MSi corresponding to 150 and 300 mg of Si per kg of soil, 0Zn with no Zn addition, and 3Zn equivalent to 3 mg of Zn per kg of soil, under two SWC: 0.4 field capacity (FC) representing drought stress, and 0.8 FC representing optimal moisture. Treatments marked with different letters differ significantly according to Duncan's multiple range test, with significance set at $p \leq 0.05$



3.7 Influence of Experimental Treatments On Root Hair Growth and Rhizosheath Development

In the untreated control, root hair characteristics did not differ significantly between plants grown in plain water and those subjected to 20% PEG to simulate osmotic stress (Fig. 7a and b). Application of 2 mM Si increased root hair length by 76.59% under water conditions and by 107.91% under PEG-induced stress. Likewise, 2 mM Zn application increased root hair length by 69.30% and 141.46% under water and PEG conditions, respectively (Fig. 7a). Root hair density followed a similar pattern: Si application elevated root hair density by 42.11% in water and by 65.14% under

PEG stress, while Zn application increased it by 57.89% in water and 42.28% under PEG conditions (Fig. 7b).

Rhizosheath formation was markedly influenced by SWC and Zn application, as well as by the interactive effect of Si and SWC (Fig. 7c–e). A reduction in SWC from 0.8 FC to 0.4 FC enhanced rhizosheath formation by 17.49% (Fig. 7c). Independent Zn application resulted in a 6.83% increase in rhizosheath mass (Fig. 7d). Under drought stress, applying 150 and 300 mg Si kg⁻¹ increased rhizosheath formation by 20.50% and 38.40%, respectively, while under optimal moisture, the increases were 8.16% and 13.17% (Fig. 7e).

Fig. 5 Effects of soil water content (SWC), silicon (Si), and zinc (Zn) on pH (a–d) and organic carbon (OC) content (e–h) in rhizosphere and bulk (non-rhizosphere) soils. The treatments consisted of 0Si with no Si addition, 150MSi and 300MSi corresponding to 150 and 300 mg of Si per kg of soil, 0Zn with no Zn addition, and 3Zn equivalent to 3 mg of Zn per kg of soil, under two SWC: 0.4 field capacity (FC) representing drought stress, and 0.8 FC representing optimal moisture. Treatments marked with different letters differ significantly according to Duncan's multiple range test, with significance set at $p \leq 0.05$

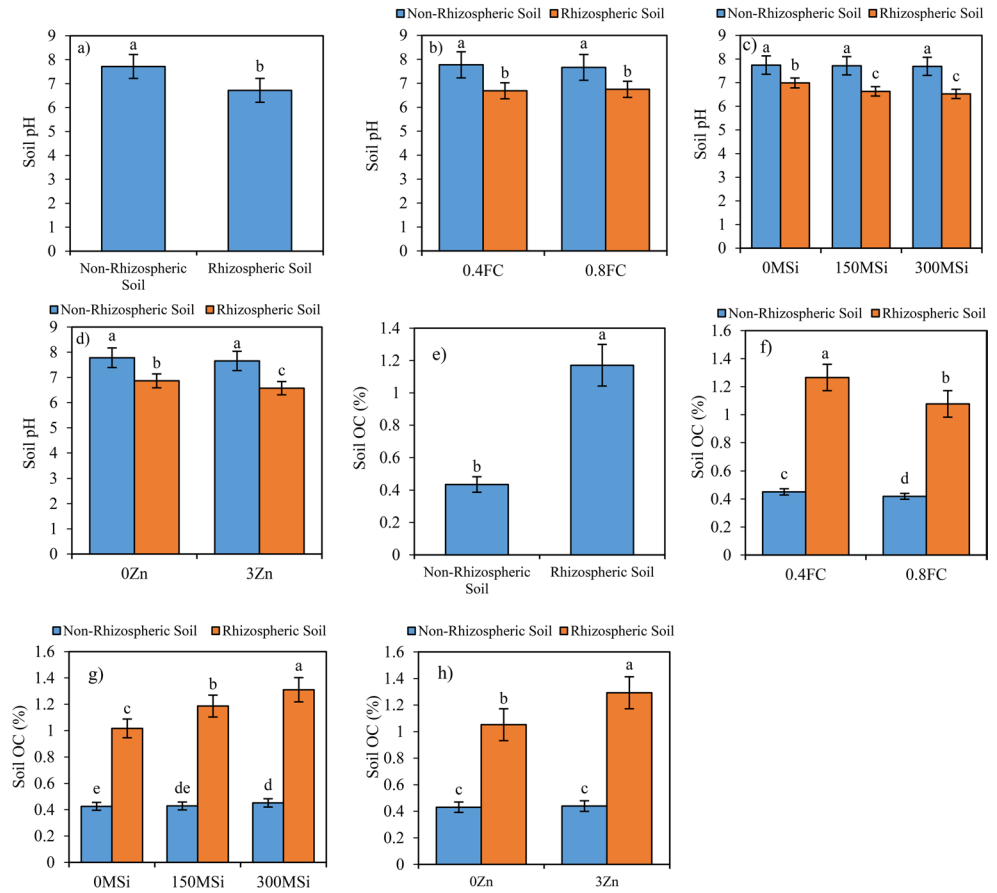
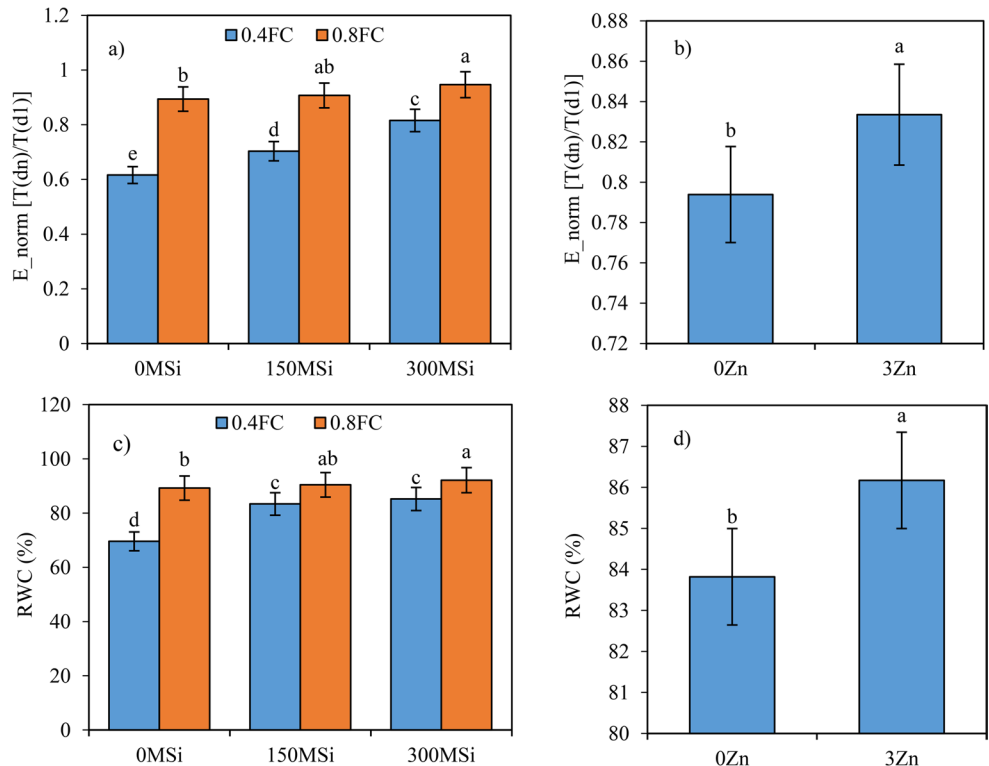


Fig. 6 Effects of the interaction between silicon (Si) and soil water content (SWC) (a, c), and the independent effect of zinc (Zn) (b, d), on normalized transpiration rate (E_{norm} , a, b) and relative leaf water content (RWC) (c, d) in wheat leaves. The treatments consisted of 0Si with no Si addition, 150MSi and 300MSi corresponding to 150 and 300 mg of Si per kg of soil, 0Zn with no Zn addition, and 3Zn equivalent to 3 mg of Zn per kg of soil, under two SWC: 0.4 field capacity (FC) representing drought stress, and 0.8 FC representing optimal moisture. Treatments marked with different letters differ significantly according to Duncan's multiple range test, with significance set at $p \leq 0.05$



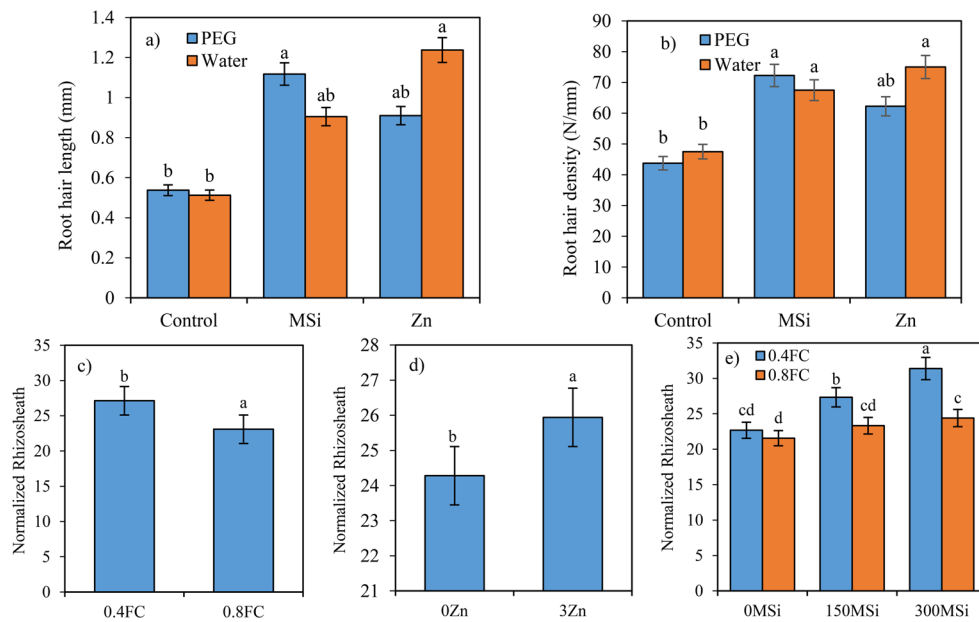


Fig. 7 Effects of silicon (Si) and zinc (Zn) on root hair length (**a**) and density (**b**) in wheat grown under hydroponic conditions, and effects of osmotic condition (**c**), independent Zn application (**d**), and the interactive effect of Si and soil water content (SWC) (**e**) on rhizosheath development per gram of root dry weight in pot-grown wheat. In panels (**a**) and (**b**), PEG denotes 20% polyethylene glycol, Si refers to 2 mM silicon, and Zn to 2 mM zinc. In panels (**c**) to (**e**), treatments

include 0Si with no Si addition, 150MSi and 300MSi corresponding to 150 and 300 mg of Si per kg of soil, 0Zn with no Zn addition, and 3Zn equivalent to 3 mg of Zn per kg of soil, under two SWC: 0.4 field capacity (FC) representing drought stress, and 0.8 FC representing optimal moisture. Treatments marked with different letters differ significantly according to Duncan's multiple range test, with significance set at $p \leq 0.05$

4 Discussion

This study presents robust evidence that drought stress substantially impairs nutrient uptake in wheat, primarily driven by reduced soil moisture and consequent alterations in nutrient availability and rhizosphere dynamics. Notably, exogenous applications of Si and Zn significantly alleviated the detrimental effects of drought by enhancing nutrient acquisition, with the greatest efficacy observed under water-limited conditions (Fig. 8). The underlying mechanisms are multifactorial: Si and Zn modulated rhizosphere chemistry through pH reduction and increased OC content, indicative of enhanced root exudation. These biochemical changes improved the solubility and bioavailability of essential nutrients in the rhizosphere.

Furthermore, Si and Zn treatments promoted key root morphological adaptations, including increased rhizosheath formation and the proliferation of longer, denser root hairs, which collectively improved hydraulic conductivity and reinforced physical soil-root interactions. These structural modifications facilitated more effective water extraction from drying soils, thereby supporting improved plant water status under drought stress. Together, these integrated physiological, biochemical, and morphological responses collectively optimize water and nutrient uptake, highlighting

the critical roles of Si and Zn in mitigating drought-induced nutrient limitations in wheat (Fig. 8).

4.1 Soil Drying and Nutrient Uptake

This study clearly demonstrated that drought stress significantly reduced the concentrations of essential macronutrients (P, K, S) and micronutrients (Fe, Zn, Cu, Mn) in wheat leaves (Figs. 1 and 2). This reduction can be attributed to several interacting mechanisms. Under water deficit, both root tissues and surrounding soil undergo volumetric contraction, leading to the formation of discontinuities or air gaps at the root–soil interface (Liu et al. 2015; North and Nobel 1997). This physical disruption markedly increases the hydraulic resistance at the root–soil interface, lowering the capacity for both water and nutrient flux. These gaps disrupt hydraulic and physical continuity, which is crucial for effective water and nutrient uptake, thus limiting root access to soil solution (Cheraghi et al. 2023b). Correspondingly, our physiological data showed a marked decrease in RWC, accompanied by stomatal closure and reduced transpiration rates (Fig. 6a–d). The decline in leaf water potential under drought triggers abscisic acid (ABA) signaling, leading to stomatal closure (Kane and McAdam 2023). While this conserves water, the resulting suppression of transpiration-driven mass flow restricts the delivery of mobile nutrients

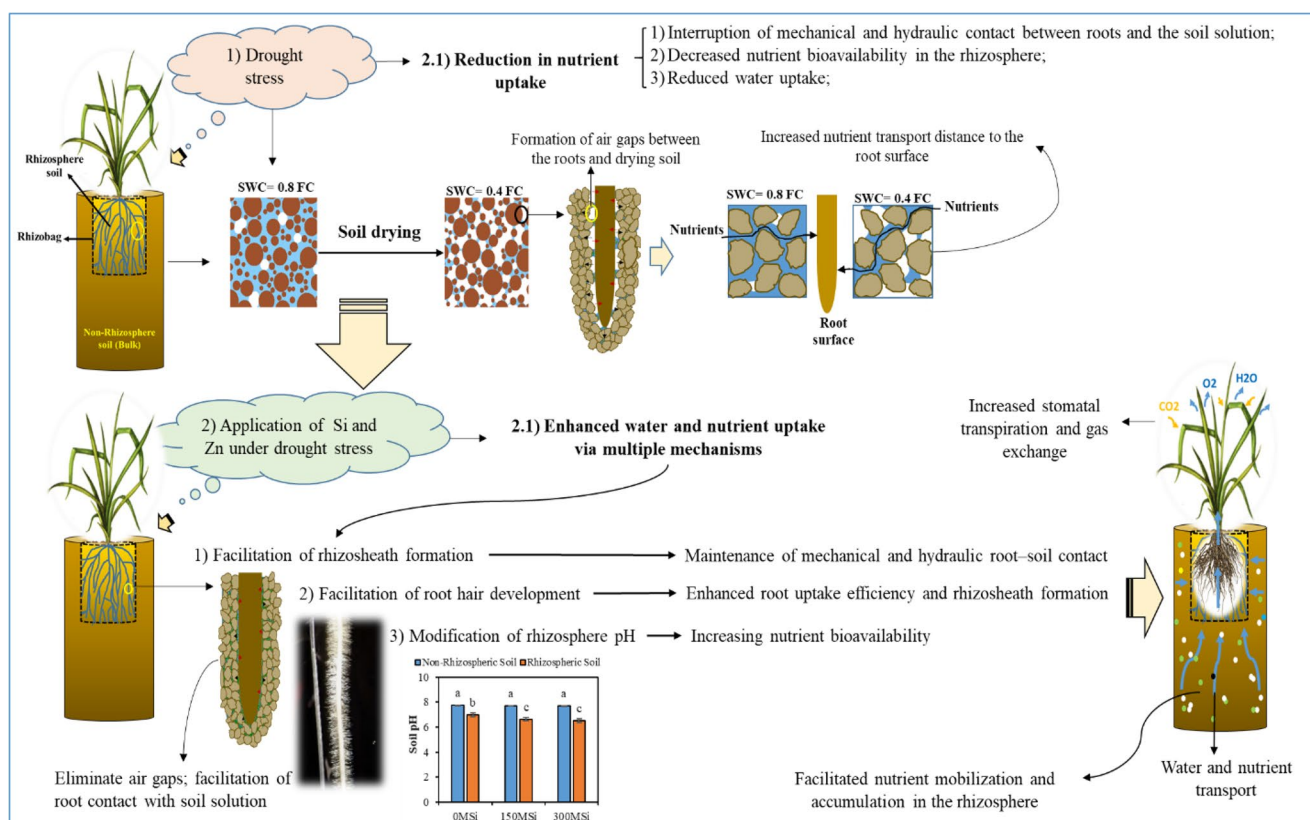


Fig. 8 Illustrated mechanism highlighting how silicon (Si) and zinc (Zn) enhance water and nutrient absorption in wheat exposed to drought conditions

to the root surface, thereby exacerbating nutrient shortages under drought stress.

In addition, soil moisture critically regulates nutrient bioavailability and mobility. When soils dry, the reduced water-filled pore space not only hinders nutrient diffusion but also diminishes the chemical potential gradient driving nutrient movement toward roots, directly impairing nutrient supply (Marschner and Rengel 2023). Drought conditions enhance nutrient fixation through increased adsorption to mineral and organic soil particles and promote ion precipitation, collectively decreasing nutrient concentrations in the soil solution (Ahanger et al. 2016; Huntley 2023). Drought-induced shifts in soil pH and redox potential can convert soluble forms of micronutrients such as Fe^{2+} and Mn^{2+} into insoluble oxides, further limiting their bioavailability (Xu et al. 2021). Importantly, microbial activity, which is vital for nutrient cycling and mineralization, declines sharply during drought (Li et al. 2024). Our data corroborate these mechanisms, revealing significant reductions in rhizospheric phosphorus, K, Fe, and Zn concentrations under drought stress (Figs. 3 and 4), demonstrating how intertwined physical, chemical, and biological changes converge to constrain nutrient uptake efficiency.

Interestingly, despite overall nutrient reductions, foliar concentrations of Ca, Mg, Si, and B were elevated in drought-stressed plants (Figs. 1i–l and 2j). This enrichment is partly due to a concentration effect caused by reduced biomass accumulation under water stress (results not reported), resulting in nutrient partitioning into a smaller tissue volume (Cooke and Carey 2023). Beyond this passive effect, these elements have crucial physiological roles in drought tolerance: Ca stabilizes cellular membranes and mediates intracellular signaling (Hepler 2005); Mg is essential for chlorophyll synthesis and photosynthetic function (Hermans et al. 2013); B maintains cell wall integrity and facilitates sugar transport (Camacho-Cristóbal et al. 2008); and Si enhances water retention, regulates stomatal behavior, and activates antioxidant defenses (Ma and Yamaji 2006). Plants may also actively upregulate specific transporters, such as *Lsi* for Si and *BOR1* for B, to increase uptake and retention during drought (Hosseini et al. 2017; Yoshinari et al. 2021). Furthermore, drought-induced changes in transpiration and xylem sap flow influence nutrient redistribution within the plant (Bahrun et al. 2002). Overall, these findings suggest that elevated leaf Ca, Mg, Si, and B concentrations result from both passive concentration due to growth inhibition

and active physiological adjustments that support drought adaptation.

4.2 Role of Si and Zn in Enhancing Nutrient Uptake Under Drought Stress

The findings of this study demonstrate that exogenous application of Si and Zn, individually or in combination, significantly improved leaf nutrient concentrations in wheat, with more pronounced effects under drought conditions. Independent application of Si increased the concentrations of N, P, K, Cu, Mn, and B, while the interactive effect of Si and SWC notably elevated levels of Ca, Fe, and Si in leaf tissues (Figs. 1 and 2). Likewise, Zn application alone enhanced N, Si, Fe, Zn, and Cu concentrations but reduced Mn under well-watered conditions. The interaction between Zn and SWC significantly increased Ca, Zn, and B levels. Notably, co-application of Si and Zn resulted in significant combined increases in S, Mg, and Zn, particularly under drought stress, highlighting the complementary roles of these nutrients in mitigating water-deficit-induced nutrient limitations. Although the statistical interaction terms ($\text{Si} \times \text{Zn} \times \text{SWC}$) were not significant for most nutrients (Table 1), the additive effects of Si and Zn were particularly beneficial under water-limited conditions.

These improvements in nutrient accumulation are attributed to a combination of rhizospheric, physiological, and molecular mechanisms. First, both Si and Zn enhanced nutrient availability in the rhizosphere, particularly under drought stress, as reflected by reductions in rhizosphere pH and increased OC content (Figs. 3, 4 and 5). The observed rhizosphere acidification likely results from a combination of enhanced organic acid exudation—indirectly stimulated by improved photosynthetic carbon allocation to roots—and increased activity of plasma membrane H^+ -ATPases, which actively extrude protons into the rhizosphere (Liang et al. 2006; Sattar et al. 2021). These changes, together with increased OC content—which likely results from a combination of enhanced root exudation, increased microbial activity, and root turnover—facilitate the mobilization of otherwise immobilized nutrients like P, Fe, and Cu through chelation and pH-driven desorption (Boldt-Burisch et al. 2019). Second, improved plant water status—evidenced by elevated RWC and transpiration rates (Fig. 6a–d)—likely promoted nutrient transport via mass flow and diffusion (Alaoui et al. 2022). Third, Si and Zn treatments stimulated root morphological traits that increased soil exploration and nutrient acquisition, including elongation and densification of root hairs (Fig. 7e–f) and increased rhizosheath formation (Fig. 7g–i), which collectively enhanced the effective root–soil contact area (Opoku et al. 2022).

On a physiological and molecular level, Si is known to stabilize cellular membranes, improve transpiration efficiency, and regulate nutrient transporters such as nitrate transporters (*NRTs*) and plasma membrane H^+ -ATPases (Cheraghi et al. 2023a; Liang et al. 2006). Zinc, on the other hand, facilitates nutrient uptake through the stimulation of auxin synthesis and activation of phosphatases, especially under P-deficient conditions (Lv et al. 2024; Wang and Yang 2021). The observed decline in Mn under Zn application is consistent with known ionic competition between Zn and Mn at shared transporter sites (Fan et al. 2021). Importantly, under combined Si and Zn application, additive effects were observed that included stronger rhizospheric acidification, enhanced antioxidant defense, and improved ionic homeostasis in plant tissues (Mapodzeke et al. 2021; Sattar et al. 2023). These findings underscore the potential of integrated Si and Zn supplementation as an effective agronomic strategy to overcome drought-induced nutrient limitations through both soil-mediated and plant-mediated pathways.

4.3 Leaf Nutrient Concentrations Vs. Total Plant Uptake

It is important to acknowledge a limitation of this study regarding the interpretation of nutrient uptake. Our conclusions are based exclusively on leaf nutrient concentrations, which reflect the net result of root uptake, translocation, and shoot accumulation. Under drought stress, plants often restrict the translocation of nutrients from roots to shoots as a physiological adaptation (Basu et al. 2016; Hu and Schmidhalter 2005). Therefore, the observed increases in leaf nutrient concentrations following Si and Zn application could arise from two distinct processes: (i) enhanced total uptake by the root system, or (ii) facilitated translocation from roots to shoots without a corresponding increase in total uptake.

Without data on root nutrient concentrations, it is not possible to distinguish between these mechanisms. This distinction is important because improved translocation alone may not reflect a true increase in the plant's ability to acquire nutrients from soil, but rather a redistribution of existing nutrient pools. Future studies should include measurements of root nutrient concentrations and, ideally, total plant nutrient content to fully elucidate whether Si and Zn enhance net uptake or primarily affect internal partitioning. Despite this limitation, the consistent increases in leaf nutrient concentrations, together with the observed improvements in rhizosphere nutrient availability, root morphology, and plant water status, strongly suggest that Si and Zn positively influenced the overall nutrient acquisition process under drought stress. These findings underscore the potential of integrated Si and Zn supplementation as an effective

agronomic strategy to overcome drought-induced nutrient limitations through both soil-mediated and plant-mediated pathways.

4.4 Rhizospheric Alterations and Improved Nutrient Dynamics At the Root–soil Interface

A major mechanism underlying the enhanced nutrient uptake observed under Si and Zn treatments was the improvement in nutrient bioavailability at the root–soil interface (Figs. 3 and 4). One of the most critical factors influencing nutrient solubility and mobility in soil is pH, with maximum nutrient availability generally occurring between pH 6.0 and 7.0, and an optimal point around pH 6.5 (Penn and Camberato 2019). In this study, Si application resulted in a significant reduction in rhizospheric pH by approximately 0.35–0.47 units, while Zn led to a decrease of up to 0.29 units (Fig. 5a–d). These shifts effectively lowered the soil pH from ~7.5 (non-rhizospheric baseline) to ~6.5 (in rhizosphere), creating a chemically favorable microenvironment for nutrient solubilization and uptake. This magnitude of pH change in a buffered soil system is substantial and warrants a mechanistic explanation. Importantly, this effect cannot be attributed to the direct chemical properties of the applied Si or Zn sources—monosilicic acid is a weak acid and zinc sulfate is pH-neutral—but rather reflects indirect, plant-mediated responses.

Several complementary mechanisms may underlie this rhizosphere acidification. First, Si and Zn application improved photosynthetic activity and carbon assimilation under drought stress (as evidenced by increased RWC and transpiration rates; Fig. 6a–d), which likely enhanced the availability of photoassimilates for root metabolic processes (Cheraghi et al. 2020; Coskun et al. 2019; Sattar et al. 2021). This, in turn, could stimulate the exudation of organic compounds, including organic acids such as citrate and malate, which contribute to rhizosphere acidification and nutrient mobilization (Kruthika et al. 2024). Although direct quantification of root exudates in soil is challenging, the significant increase (16–28%) in rhizospheric total organic carbon under Si and Zn treatments (Fig. 5g–h) may partly reflect enhanced root exudation (Coskun et al. 2019), although it is important to acknowledge that rhizosphere OC is a composite pool that also includes microbial biomass, sloughed-off root cells, and root turnover. Therefore, while increased OC is consistent with stimulated exudation, it cannot be exclusively attributed to this process.

Second, and more specifically, Si and Zn may stimulate the activity of plasma membrane H^+ -ATPases in root cells. These proton pumps are responsible for acidifying the rhizosphere by extruding H^+ into the apoplast, a process that is energetically coupled to ATP availability. Improved

plant energy status under Si and Zn supply—resulting from enhanced photosynthesis and reduced oxidative stress—could sustain higher H^+ -ATPase activity (Liang et al. 2006; Sattar et al. 2021). Additionally, Zn plays a critical role in maintaining membrane integrity and regulating transporter proteins (Cakmak 2000), which may indirectly support proton pump function.

We acknowledge that our experimental design was not specifically optimized to dissect these mechanisms, and the observed pH reduction was an unexpected but consistent finding. Therefore, we present these mechanistic interpretations as hypotheses supported by our data and existing literature, rather than definitive conclusions. Future studies employing targeted approaches—such as direct quantification of organic acid exudation, measurement of H^+ -ATPase activity, or gene expression analysis of proton pump isoforms—are needed to elucidate the precise pathways involved.

In addition to these plant-mediated rhizosphere modifications, Si and Zn may also directly influence nutrient dynamics through physicochemical interactions at soil exchange sites. Elevated Si levels can replace phosphate ions at adsorption sites on Fe and Al oxides, thereby enhancing P availability in the soil solution (Reithmaier et al. 2017; Schaller et al. 2019). Similarly, Zn may displace or mobilize other micronutrients, such as Fe, Mn, and Cu, through ionic competition at cation exchange sites, particularly under low-pH conditions (Selim 2016). These interactions can be further enhanced by the formation of soluble chelates and complexes in the altered rhizospheric environment. Taken together, these results reveal that Si and Zn initiate a cascade of physicochemical processes at the root–soil interface—including pH modulation, increased root exudation, and ion-exchange dynamics—that collectively improve nutrient mobility and bioavailability. Such rhizospheric adjustments under drought conditions represent a key mechanism through which Si and Zn co-application supports enhanced nutrient acquisition in wheat, reinforcing their integrative role in alleviating drought-induced nutrient limitations.

4.5 Improved Soil–plant–water Relations and Facilitation of Water and Nutrient Uptake

The ameliorative effects of Si and Zn on nutrient acquisition under drought stress were closely linked to their capacity to improve soil–plant–water relations and reinforce the plant's ability to absorb water. During drought, gradual soil drying breaks the hydraulic and physical connection between roots and surrounding soil, mainly because air pockets form at the root–soil interface (Carminati et al. 2009). This detachment reduces root contact with the soil solution, thereby limiting water uptake and the transport of mobile nutrients

(Carminati et al. 2017a). Therefore, any intervention that stabilizes root–soil connectivity has the potential to mitigate drought-induced limitations on nutrient acquisition. Our findings demonstrated that both Si and Zn significantly improved plant water status under drought. Silicon application resulted in a 14–32% increase in transpiration rate and a 19–22% enhancement in RWC, while Zn application induced respective increases of 5% and 2.8% (Fig. 6a–d). These improvements in water status likely facilitated nutrient acquisition through enhanced mass flow and diffusion-driven transport mechanisms (Zarebanadkouki et al. 2013).

A pivotal mechanism underlying this enhancement was the stimulation of rhizosheath development. The rhizosheath—a cohesive layer of soil particles bound to the root surface via root exudates and microbial metabolites—acts as a hydraulic buffer and plays a critical role in maintaining root hydration under drought stress (Cheraghi et al. 2023b; Hosseini et al. 2024). Our data revealed that Si application increased rhizosheath formation by 20–38%, while Zn enhanced it by 6.8% (Fig. 7h–i). This structural adaptation significantly improved the physical interface between roots and soil, thereby preserving water flow and facilitating the acquisition of soil-bound nutrients (Steiner et al. 2024). In addition, the rhizosheath may provide a favorable microhabitat for rhizosphere microorganisms, many of which are potentially involved in nutrient mobilization processes, such as phosphate solubilization and siderophore production (Etesami 2021). Although microbial parameters were not directly measured in this study, this represents a plausible pathway that warrants further investigation.

The formation and functionality of the rhizosheath are governed by a suite of biophysical and physiological traits, including soil moisture fluctuations, root hair length and density, and root exudate composition—particularly mucilage (Aslam et al. 2022; Cheraghi et al. 2023b). In accordance with this framework, our study showed that rhizosheath formation was 17.4% greater under drought stress compared to optimal moisture conditions (Fig. 7g). Notably, Si and Zn treatments significantly enhanced root hair elongation and density (Fig. 7e–f), which are essential for anchoring soil particles and stabilizing the rhizosheath structure. These effects were further supported by increased rhizospheric OC content (Fig. 5g–h), suggesting a stimulation of root exudation under Si and Zn supply. Root exudates, not only enhance rhizosheath cohesion but also contribute to the mobilization of otherwise inaccessible nutrients by altering rhizosphere chemistry (Kruthika et al. 2024; Zambelli et al. 2025). These changes collectively strengthened the hydraulic continuity and nutrient accessibility of the rhizosphere, ultimately enabling more efficient water and nutrient uptake under drought conditions.

4.6 Root Hair Development and Facilitation of Nutrient Uptake

Enhanced root hair elongation and density represent a critical morphological adaptation, increasing root–soil contact and improving water and nutrient uptake under drought conditions (Carminati et al. 2009, 2017b). In the present study, the exogenous application of Si and Zn significantly promoted root hair development in wheat, with increases of 76–142% in length and 42–65% in density under both hydroponic and PEG-induced drought conditions (Fig. 7e–f). These changes were positively associated with improved nutrient concentrations in leaf tissues, suggesting that root hair proliferation played a key role in enhancing nutrient uptake efficiency.

The stimulatory effects of Si on root hair growth can be attributed to several molecular and physiological mechanisms. First, Si influences the cell cycle by raising the fraction of meristematic root cells in the G1 and S phases and supporting the transition from G2 to M phase, thereby enhancing cell division and elongation (Tripathi et al. 2012, 2022a). Second, Si improves cellular redox balance by enhancing glutathione metabolism and increasing the GSH/GSSG ratio, thereby controlling ROS levels and activating ROS-dependent signaling pathways involved in root hair development (Herschbach et al. 2010). Third, Si elevates auxin concentrations in root tissues by upregulating genes responsible for auxin biosynthesis and transport, such as YUCCA and PIN family members, which are crucial for the initiation and elongation of root hairs (Tripathi et al., 2022a, b). Additionally, Si-induced improvements in photosynthetic activity and carbon assimilation may increase carbon allocation to the roots, further supporting the energetic demands of root hair formation and cell wall expansion (Zhou et al. 2023).

Zinc application also contributes to enhanced root hair development via complementary mechanisms. Zn serves as a cofactor in the enzymatic synthesis of auxin via tryptophan-dependent pathways, leading to increased indole-3-acetic acid production in roots and promoting hair initiation (Castillo-González et al. 2018; Wang and Yang 2021). It also activates cell wall-modifying enzymes such as expansins and pectin methylesterases, which facilitate cell wall loosening and tip growth of root hairs (Castillo-González et al. 2018; Gate et al. 2024). Moreover, Zn helps maintain membrane integrity and limits oxidative damage (e.g., via Cu/Zn-SOD and reduced lipid peroxidation), thereby sustaining elongation of rapidly growing cells, including root hairs, under drought (Cakmak 2000; Umair Hassan et al. 2020). Together, these Zn-mediated mechanisms increase root hair initiation and length, improving the physical interface with soil.

It is important to acknowledge that these root hair measurements were obtained from a short-term hydroponic experiment using PEG to simulate osmotic stress. While hydroponic systems enable precise phenotyping of root hairs—which remains technically challenging in soil-based systems—they do not fully capture the physical complexity of the soil environment, including mechanical impedance, soil aggregation, and particle–root interactions. Therefore, the observed trends in root hair elongation and density should be viewed as indicative of the potential of Si and Zn to stimulate these traits under osmotic stress, rather than direct quantitative predictions of root hair phenotypes in soil. Nonetheless, the consistency between the hydroponic observations and the increased rhizosheath formation and nutrient uptake in the soil experiment supports the biological relevance of these findings.

5 Conclusion

This study provides compelling evidence that exogenous application of Si and Zn can substantially improve nutrient acquisition in wheat exposed to variable soil water conditions, particularly under drought stress, as evidenced by increased leaf nutrient concentrations. Drought strongly constrained nutrient bioavailability and uptake by altering rhizosphere chemistry and weakening soil–root hydraulic continuity. However, Si and Zn alleviated these limitations through complementary mechanisms: both reduced rhizosphere pH, stimulated the secretion of carbon-rich exudates, and enhanced the solubility and mobility of macro- and micronutrients. In parallel, they promoted key root morphological traits—such as longer and denser root hairs and greater rhizosheath development—that improved mechanical and hydraulic contact with drying soil, thereby sustaining water uptake and facilitating nutrient transport via mass flow. Beyond their direct effects on nutrient dynamics, Si and Zn also contributed to improved plant water relations, as indicated by higher RWC and transpiration under drought stress. Collectively, these findings highlight the dual role of Si and Zn in simultaneously modulating rhizosphere chemistry and root architecture, which together underpin enhanced nutrient use efficiency and drought resilience in wheat. From an agronomic perspective, integrating Si and Zn into nutrient management strategies holds considerable potential to strengthen the sustainability of wheat production in water-limited environments. Future research should focus on elucidating the molecular and biochemical pathways governing Si- and Zn-mediated regulation of root exudation, root hair development, and nutrient mobilization under progressive drought, as well as the potential involvement of rhizosphere microorganisms in these processes.

Validating these mechanisms under field conditions and across different wheat genotypes also remains a priority.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s42729-026-03337-y>.

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Data Availability Data will be made available on request.

Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

References

- Ahanger MA, Morad-Talab N, Abd-Allah EF, Ahmad P, Hajiboland R (2016) Plant growth under drought stress: Significance of mineral nutrients. *Water stress crop plants: sustainable approach* 2:649–668. <https://doi.org/10.1002/9781119054450.ch37>
- Alaoui I, Serbouti S, Ahmed H, Mansouri I, El Kamari F, Taroq A, Ousaad D, Squalli W, Farah A (2022) The mechanisms of absorption and nutrients transport in plants: A review. *Trop J Nat Prod Res* 6:8–14 Available online at: <https://www.tjnpr.org>
- Ashfaq W, Fuentes S, Brodie G, Gupta D (2022) The role of silicon in regulating physiological and biochemical mechanisms of contrasting bread wheat cultivars under terminal drought and heat stress environments. *Front Plant Sci* 13:955490. <https://doi.org/10.3389/fpls.2022.955490>
- Aslam MM, Karanja JK, Dodd IC, Waseem M, Weifeng X (2022) Rhizosheath: An adaptive root trait to improve plant tolerance to phosphorus and water deficits? *Plant Cell Environ* 45:2861–2874. <https://doi.org/10.1111/pce.14395>
- Bahrn A, Jensen CR, Asch F, Mogensen VO (2002) Drought-induced changes in xylem pH, ionic composition, and ABA concentration act as early signals in field-grown maize (*Zea mays* L.). *J Exp Bot* 53:251–263. <https://doi.org/10.1093/jexbot/53.367.251>
- Basu S, Ramegowda V, Kumar A, Pereira A (2016) Plant adaptation to drought stress. *F1000Res* 5:F1000 Faculty Rev-1554. <https://doi.org/10.12688/f1000research.7678.1>
- Boldt-Burisch K, Schneider BU, Naeth MA, Huettl RF (2019) Root exudation of organic acids of herbaceous pioneer plants and their growth in sterile and non-sterile nutrient-poor, sandy soils from post-mining sites. *Pedosphere* 29:34–44. [https://doi.org/10.1016/S1002-0160\(18\)60056-6](https://doi.org/10.1016/S1002-0160(18)60056-6)

- Broadley MR, White PJ, Hammond JP, Zelko I, Lux A (2007) Zinc in plants. *New Phytol* 173:677–702. <https://doi.org/10.1111/j.1469-8137.2007.01996.x>
- Cakmak I (2000) Tansley Review 111 Possible roles of zinc in protecting plant cells from damage by reactive oxygen species. *New Phytol* 146:185–205. <https://doi.org/10.1046/j.1469-8137.2000.00630.x>
- Camacho-Cristóbal JJ, Rexach J, González-Fontes A (2008) Boron in plants: deficiency and toxicity. *J Integr Plant Biol* 50:1247–1255. <https://doi.org/10.1111/j.1744-7909.2008.00742.x>
- Carminati A, Vetterlein D, Weller U, Vogel H-J, Oswald SE (2009) When roots lose contact. *Vadose Zone J* 8:805–809. <https://doi.org/10.2136/vzj2008.0147>
- Carminati A, Benard P, Ahmed MA, Zarebanadkouki M (2017a) Liquid bridges at the root-soil interface. *Plant Soil* 417:1–15. <https://doi.org/10.1007/s11104-017-3227-8>
- Carminati A, Passioura JB, Zarebanadkouki M, Ahmed MA, Ryan PR, Watt M, Delhaize E (2017b) Root hairs enable high transpiration rates in drying soils. *New Phytol* 216:771–781. <https://doi.org/10.1111/nph.14715>
- Castiglione MR, Bottega S, Sorce C, Spanò C (2023) Effects of zinc oxide particles with different sizes on root development in *Oryza sativa*. *Rice Sci* 30:449–458. <https://doi.org/10.1016/j.rsci.2023.03.016>
- Castillo-González J, Ojeda-Barrios D, Hernández-Rodríguez A, González-Franco AC, Robles-Hernández L, López-Ochoa GR (2018) Zinc metalloenzymes in plants. *Interciencia* 43:242–248
- Ceolin S (2024) *Root Dynamics Under Varying Water and Nutrient Supply* [Doctoral thesis, Unilu - University of Luxembourg]. ORBilu-University of Luxembourg. <https://orbilu.uni.lu/handle/10993/61739>
- Cheraghi M, Moteszarezhadeh B, Alikhani HA (2020) Nutritional and morpho-physiological responses of tomato plant (*Lycopersicon esculentum* Mill) affected by biological and chemical fertilizers. *Iran J Soil Water Res* 51:2559–2574. <https://doi.org/10.22059/ijswr.2020.303408.668631>
- Cheraghi M, Moteszarezhadeh B, Mousavi SM, Ma Q, Ahmadabadi Z (2023a) Silicon (Si): A regulator nutrient for optimum growth of wheat under salinity and drought stresses-a review. *J Plant Growth Regul* 42:5354–5378. <https://doi.org/10.1007/s00344-023-10959-4>
- Cheraghi M, Mousavi SM, Zarebanadkouki M (2023b) Functions of rhizosphere on facilitating the uptake of water and nutrients under drought stress: A review. *Plant Soil* 1–25. <https://doi.org/10.1007/s11104-023-06126-z>
- Cheraghi M, Moteszarezhadeh B, Mousavi SM, Basirat M, Alikhani HA, Zarebanadkouki M (2024) Application of silicon improves rhizosphere formation, morpho-physiological and biochemical responses of wheat under drought stress. *Plant Soil*. <https://doi.org/10.1007/s11104-024-06584-z>
- Cheraghi M, Moteszarezhadeh B, Mousavi SM, Basirat M, Alikhani HA (2025) Phosphorus bioavailability and silicon fractionation in wheat rhizosphere affected by soil water content and silicon application. *Rhizosphere* 33:101017. <https://doi.org/10.1016/j.rhisph.2025.101017>
- Cooke J, Carey JC (2023) Stress alters the role of silicon in controlling plant water movement. *Funct Ecol* 37:2985–2999. <https://doi.org/10.1111/1365-2435.14447>
- Coskun D, Deshmukh R, Sonah H, Menzies JG, Reynolds O, Ma JF, Kronzucker HJ, Bélanger RR (2019) The controversies of silicon's role in plant biology. *New Phytol* 221:67–85. <https://doi.org/10.1111/nph.15343>
- Dane JH, Topp CG (2020) *Methods of soil analysis, Part 4: Physical methods*. John Wiley & Sons, New York, NY
- Delhaize E, James RA, Ryan PR (2012) Aluminium tolerance of root hairs underlies genotypic differences in rhizosphere size of wheat (*Triticum aestivum*) grown on acid soil. *New Phytol* 195:609–619. <https://doi.org/10.1111/j.1469-8137.2012.04183.x>
- Etesami H (2021) Potential advantage of rhizosphere microbiome, in contrast to rhizosphere microbiome, to improve drought tolerance in crops. *Rhizosphere* 20:100439. <https://doi.org/10.1016/j.rhisph.2021.100439>
- Fan X, Zhou X, Chen H, Tang M, Xie X (2021) Cross-talks between macro-and micronutrient uptake and signaling in plants. *Front Plant Sci* 12:663477. <https://doi.org/10.3389/fpls.2021.663477>
- Gate T, Hill L, Miller AJ, Sanders D (2024) AtLAR1 is a Zn transporter that regulates auxin metabolism in *Arabidopsis thaliana*. *J Exp Bot* 75:1437–1450. <https://doi.org/10.1093/jxb/erad468>
- Georgiadis A, Rinklebe J, Straubinger M, Rennert T (2017) Silicon fractionation in mollic fluvisols along the central Elbe River, Germany. *Catena* 153:100–105. <https://doi.org/10.1016/j.catena.2017.01.027>
- Hepler PK (2005) Calcium: a central regulator of plant growth and development. *Plant Cell* 17:2142–2155. <https://doi.org/10.1105/tpc.105.032508>
- Hermans C, Conn SJ, Chen J, Xiao Q, Verbruggen N (2013) An update on magnesium homeostasis mechanisms in plants. *Metallomics* 5:1170–1183. <https://doi.org/10.1039/c3mt20223b>
- Herschbach C, Scheerer U, Rennenberg H (2010) Redox states of glutathione and ascorbate in root tips of poplar (*Populus tremula* × *P. alba*) depend on phloem transport from the shoot to the roots. *J Exp Bot* 61:1065–1074. <https://doi.org/10.1093/jxb/erp371>
- Horneck DA, Miller RO (2019) Determination of total nitrogen in plant tissue. *Handbook of reference methods for plant analysis*. CRC, pp 75–83
- Hosseini SA, Maillard A, Hajirezaei MR, Ali N, Schwarzenberg A, Jamois F, Yvin J-C (2017) Induction of barley Silicon transporter HvLsi1 and HvLsi2, increased silicon concentration in the shoot and regulated starch and ABA homeostasis under osmotic stress and concomitant Potassium deficiency. *Front Plant Sci* 8:1359. <https://doi.org/10.3389/fpls.2017.01359>
- Hosseini B, Cheraghi M, Hiesch S, Yu P, Zarebanadkouki M (2024) Contrasting rhizosphere formation capacities in two maize inbred lines: implications for water and nutrient uptake. *Plant Soil*. <https://doi.org/10.1007/s11104-024-06883-5>
- Hu Y, Schmidhalter U (2005) Drought and salinity: a comparison of their effects on mineral nutrition of plants. *J Plant Nutr Soil Sci* (1999) 168:541–549. <https://doi.org/10.1002/jpln.200420516>
- Huntley BJ (2023) *Soil, water and nutrients. Ecology of Angola: Terrestrial biomes and ecoregions*. Springer, pp 127–147. https://doi.org/10.1007/978-3-031-18923-4_6
- Kane C, McAdam S (2023) Absciscic acid driven stomatal closure during drought in anisohydric *Fagus sylvatica*. *J Plant Hydraul*. <https://doi.org/10.20870/jph.2023.002>
- Korndörfer GH, Coelho NM, Snyder GH, Mizutani CT (1999) Avaliação de métodos de extração de silício em solos cultivados com arroz de sequeiro. *Rev Bras Cienc Solo* 23:101–106. <https://doi.org/10.1590/s0100-06831999000100013>
- Kruthika S, Ashu A, Anand A, Sammi Reddy K, Vara Prasad PV, Gurumurthy S (2024) Unveiling the role of root exudates in plant adaptation to drought and heat stress. *J Crop Health*. <https://doi.org/10.1007/s10343-024-01013-8>
- Li P, Sun Y, Hou X (2024) Drought stress impacts soil microbial nutrient limitation more strongly than O₃ pollution. *Appl Soil Ecol* 201:105496. <https://doi.org/10.1016/j.apsoil.2024.105496>
- Liang Y, Zhang W, Chen Q, Liu Y, Ding R (2006) Effect of exogenous silicon (Si) on H⁺-ATPase activity, phospholipids and fluidity of plasma membrane in leaves of salt-stressed barley (*Hordeum vulgare* L.). *Environ Exp Bot* 57:212–219. <https://doi.org/10.1016/j.envexpbot.2005.05.012>
- Liu XP, Zhang WJ, Wang XY, Cai YJ, Chang JG (2015) Root-soil air gap and resistance to water flow at the soil-root interface of

- Robinia pseudoacacia*. Tree Physiol 35:1343–1355. <https://doi.org/10.1093/treephys/tpv075>
- Lv H, Ding J, Zhang L, Wang C, Cai H (2024) Zinc application facilitates the turnover of organic phosphorus in rice rhizosphere soil by modifying microbial communities. Plant Soil 498:77–92. <https://doi.org/10.1007/s11104-022-05810-w>
- Ma JF, Yamaji N (2006) Silicon uptake and accumulation in higher plants. Trends Plant Sci 11:392–397. <https://doi.org/10.1016/j.plants.2006.06.007>
- Mapodzeke JM, Adil MF, Wei D, Joan HI, Ouyang Y, Shamsi IH (2021) Modulation of key physio-biochemical and ultrastructural attributes after synergistic application of zinc and silicon on rice under cadmium stress. Plants (Basel) 10:87. <https://doi.org/10.3390/plants10010087>
- Marschner P, Rengel Z (2023) Nutrient availability in soils. Marschner's Mineral Nutrition of Plants. Elsevier, pp 499–522. <https://doi.org/10.1016/B978-0-12-819773-8.00003-4>
- Mostofa MG, Rahman MM, Ansary MMU, Keya SS, Abdelrahman M, Miah MG, Phan Tran L-S (2021) Silicon in mitigation of abiotic stress-induced oxidative damage in plants. Crit Rev Biotechnol 41:918–934. <https://doi.org/10.1080/07388551.2021.1892582>
- Neumann M, Nöske R, Bach G, Glaubauf T, Bartoszek M, Strauch P (2011) A procedure for rapid determination of the silicon content in plant materials. Z Naturforsch B J Chem Sci 66:289–294. <https://doi.org/10.1515/znB-2011-0311>
- North GB, Nobel PS (1997) Drought-induced changes in soil contact and hydraulic conductivity for roots of *Opuntia ficus-indica* with and without rhizosheaths. Plant Soil 191:249–258. <https://doi.org/10.1023/A:1004213728734>
- Olsen SR (1954) Estimation of available phosphorus in soils by extraction with sodium bicarbonate. US Department of Agriculture
- Opoku VA, Yawson DO, Asare PA, Afutu E, Kotochi MC, Amoah KK, Adu MO (2022) Root hair and rhizosheath traits contribute to genetic variation and phosphorus use efficiency in cowpea (*Vigna unguiculata* (L.) Walp). Rhizosphere 21:100463. <https://doi.org/10.1016/j.rhisph.2021.100463>
- Penn CJ, Camberato JJ (2019) A critical review on soil chemical processes that control how soil pH affects phosphorus availability to plants. Agriculture 9:120. <https://doi.org/10.3390/agriculture9060120>
- Ranjan A, Sinha R, Singla-Pareek SL, Pareek A, Singh AK (2022) Shaping the root system architecture in plants for adaptation to drought stress. Physiol Plant 174:e13651. <https://doi.org/10.1111/ppl.13651>
- Reithmaier G-M, Knorr K-H, Arnhold S, Planer-Friedrich B, Schaller J (2017) Enhanced silicon availability leads to increased methane production, nutrient and toxicant mobility in peatlands. Sci Rep 7:8728. <https://doi.org/10.1038/s41598-017-09130-3>
- Rhoades JD (1996) Salinity: Electrical conductivity and total dissolved solids. Methods soil analysis: Part 3 Chem methods 5:417–435. <https://doi.org/10.2136/sssabookser5.3.c14>
- Roychoudhry S, Kepinski S (2022) Auxin in root development. Cold Spring Harb Perspect Biol 14:a039933. <https://doi.org/10.1101/cshperspect.a039933>
- Sattar A, Wang X, Abbas T, Sher A, Ijaz M, Ul-Allah S, Irfan M, Butt M, Wahid MA, Cheema M (2021) Combined application of zinc and silicon alleviates terminal drought stress in wheat by triggering morpho-physiological and antioxidants defense mechanisms. PLoS One 16:e0256984. <https://doi.org/10.1371/journal.pone.0256984>
- Sattar A, Sher A, Ijaz M, Ul-Allah S, Hussain S, Rasheed U, Hussain J, Al-Qahtani SM, Al-Harbi NA, Mahmoud SF (2023) Modulation of antioxidant defense mechanisms and morpho-physiological attributes of wheat through exogenous application of silicon and melatonin under water deficit conditions. Sustainability 15:7426
- Schaller J, Faucherre S, Joss H, Obst M, Goeckede M, Planer-Friedrich B, Peiffer S, Gilfedder B, Elberling B (2019) Silicon increases the phosphorus availability of Arctic soils. Sci Rep 9:449. <https://doi.org/10.1038/s41598-018-37104-6>
- Seleiman MF, Al-Suhaibani N, Ali N, Akmal M, Alotaibi M, Refay Y, Dindaroglu T, Abdul-Wajid HH, Battaglia ML (2021) Drought stress impacts on plants and different approaches to alleviate its adverse effects. Plants 10:259. <https://doi.org/10.3390/plants10020259>
- Selim HM (2016) Bioavailability of micronutrients and other trace elements in soils. Bull La State Univ Agric Bull 894
- Shahbazi K, Cheraghi M, Marzi M, Hasheminasab Zavareh KS (2022) The effect of extractant type and soil/extractant ratio on the extraction of soil available potassium. Iran J Soil Water Res 53:1481–1497. <https://doi.org/10.22059/ijswr.2022.341938.669253>
- Sparks DL, Page AL, Helmke PA, Loeppert RH (2020) Methods of soil analysis, part 3: Chemical methods. Wiley, New York, NY
- Steiner FA, Wild AJ, Tyborski N, Tung S, Koehler T, Buegger F, Carminati A, Eder B, Groth J, Hesse BD (2024) Rhizosheath drought responsiveness is variety-specific and a key component of below-ground plant adaptation. New Phytol 242:479–492. <https://doi.org/10.1111/nph.19638>
- Tripathi DK, Singh VP, Kumar D, Chauhan DK (2012) Impact of exogenous silicon addition on chromium uptake, growth, mineral elements, oxidative stress, antioxidant capacity, and leaf and root structures in rice seedlings exposed to hexavalent chromium. Acta Physiol Plant 34:279–289. <https://doi.org/10.1007/S11738-011-0826-5>
- Tripathi DK, Rai P, Kandhol N, Kumar A, Sahi S, Corpas FJ, Sharma S, Singh VP (2022a) Silicon palliates chromium toxicity through the formation of root hairs in rice (*Oryza sativa*) mediated by GSH and IAA. Plant Cell Physiol 63:1943–1953. <https://doi.org/10.1093/pcp/pcac150>
- Tripathi P, Tayade R, Mun B-G, Yun B-W, Kim Y (2022b) Silicon application differentially modulates root morphology and expression of PIN and YUCCA family genes in soybean (*Glycine max* L.). Front Plant Sci 13:842832. <https://doi.org/10.3389/fpls.2022b.842832>
- Turner NC (1981) Techniques and experimental approaches for the measurement of plant water status. Plant Soil 58:339–366. <https://doi.org/10.1007/BF02180062>
- Umair Hassan M, Aamer M, Umer Chattha M, Haiying T, Shahzad B, Barbanti L, Nawaz M, Rasheed A, Afzal A, Liu Y (2020) The critical role of zinc in plants facing the drought stress. Agriculture 10:396. <https://doi.org/10.3390/agriculture10090396>
- Vadez V, Grondin A, Chenu K, Henry A, Laplaze L, Millet EJ, Carminati A (2024) Crop traits and production under drought. Nat Rev Earth Environ 5:211–225. <https://doi.org/10.1038/s43017-023-00514-w>
- Vandegheer RK, Zhao C, Cibils-Stewart X, Wuhler R, Hall CR, Hartley SE, Tissue DT, Johnson SN (2021) Silicon deposition on guard cells increases stomatal sensitivity as mediated by K⁺ efflux and consequently reduces stomatal conductance. Physiol Plant 171:358–370. <https://doi.org/10.1111/ppl.13202>
- Walkley A, Black IA (1934) An examination of the Degtjareff method for determining soil organic matter, and a proposed modification of the chromic acid titration method. Soil Sci 37:29–38. <https://doi.org/10.1097/00010694-193401000-00003>
- Wang J, Yang S (2021) Dose-dependent responses of *Arabidopsis thaliana* to zinc are mediated by auxin homeostasis and transport. Environ Exp Bot 189:104554. <https://doi.org/10.1016/j.envexpbot.2021.104554>
- Wu C, Yan S, Zhang H, Luo Y (2015) Chemical forms of cadmium in a calcareous soil treated with different levels of phosphorus-containing acidifying agents. Soil Res 53:105–111. <https://doi.org/10.1071/SR14210>

- Xu L, Dong Z, Chiniquy D, Pierroz G, Deng S, Gao C, Diamond S, Simmons T, Wipf H-L, Caddell D (2021) Genome-resolved metagenomics reveals role of iron metabolism in drought-induced rhizosphere microbiome dynamics. *Nat Commun* 12:3209. <https://doi.org/10.1038/s41467-021-23553-7>
- Yoshida S, Ohnishi Y, Kitagishi K (1962) Histochemistry of silicon in rice plant: III. The presence of cuticle-silica double layer in the epidermal tissue. *Soil Sci Plant Nutr* 8:1–5. <https://doi.org/10.1080/00380768.1962.10430982>
- Yoshinari A, Hosokawa T, Beier MP, Oshima K, Ogino Y, Hori C, Takasuka TE, Fukao Y, Fujiwara T, Takano J (2021) Transport-coupled ubiquitination of the borate transporter BOR1 for its boron-dependent degradation. *Plant Cell* 33:420–438. <https://doi.org/10.1093/plcell/koaa020>
- Zahmatkesh Y, Souhani A, Pessarakli M (2024) Physiological, biochemical, and functional changes in quinoa (*Chenopodium quinoa* Willd) under potassium and zinc applications in drought stress conditions. *J Plant Nutr* 47:3622–3645. <https://doi.org/10.1080/01904167.2024.2380783>
- Zambelli A, Nocito FF, Araniti F (2025) Unveiling the multifaceted roles of root exudates: chemical interactions, allelopathy, and agricultural applications. *Agron (Basel)* 15:1–30. <https://doi.org/10.3390/agronomy15040845>
- Zarebanadkouki M, Kim YX, Carminati A (2013) Where do roots take up water? Neutron radiography of water flow into the roots of transpiring plants growing in soil. *New Phytol* 199:1034–1044. <https://doi.org/10.1111/nph.12330>
- Zhou Z, Struik PC, Gu J, van der Putten PEL, Wang Z, Yin X, Yang J (2023) Leaf-colour modification affects canopy photosynthesis, dry-matter accumulation and yield traits in rice. *Field Crops Res* 290:108746. <https://doi.org/10.1016/j.fcr.2022.108746>

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